

Lecture 5: Unsupervised Learning
PCA, t-SNE, Clustering, Autoencoders
Phase Diagrams of Classical Spin Models without Labels

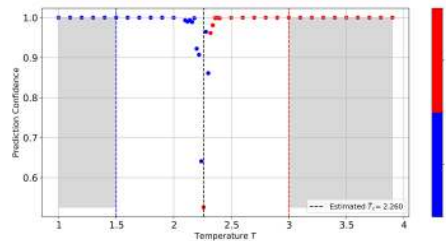
Deep Learning in Classical and Quantum Physics

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Recap: Finding T_c with Labels

- a CNN trained on spin configurations labelled "ordered" ($T < 1.5$) and "disordered" ($T > 3$)
- its average confidence is lowest at $T = 2.26$ ($L = 50$); exact value $T_c \approx 2.269$
- but the labels came from us: we knew that there are two phases, and where each of them lies



Today: what can we learn from data without labels?

1

Learning without Labels



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Unsupervised Learning

Supervised learning needs labels, and labels are expensive (every sample labelled, often by an expert), contain what we already know (the ten digits, the two phases of a magnet), and in discovery the categories are often unknown, or whether there are any.

Supervised

data: pairs $\{(\mathbf{x}_i, y_i)\}_{i=1}^n$

goal: a function $f(\mathbf{x}) \approx y$, i.e. learn $p(y \mid \mathbf{x})$

we fix the question in advance: which cell type? which phase?

Unsupervised

data: samples only, $\{\mathbf{x}_i\}_{i=1}^n$

goal: learn about $p(\mathbf{x})$, the distribution the data come from

no question fixed in advance: we want to **understand the data**

Unsupervised Methods Answer Different Questions

Understanding the data, in a broad sense:

question about the data	method
which few variables describe a sample?	dimensionality reduction
how many kinds of samples are there?	clustering
what is typical, and what is unusual?	anomaly detection
how are the samples distributed? can we draw new ones?	density estimation, generative models

The method proposes a structure; deciding what it means is ours.

High-Dimensional Data

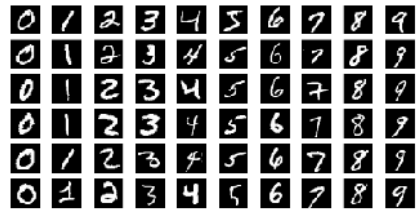
every sample is a vector $\mathbf{x} \in \mathbb{R}^D$: an MNIST digit has $D = 28 \cdot 28 = 784$ pixels, an Ising configuration $D = L^2$ spins

We cannot look at 784 dimensions. But the data do not fill $\mathbb{R}^{784}$: pick 784 random pixel values and you get noise, never a digit. Real images lie in a thin, low-dimensional part of the space.

So the analysis starts with **dimensionality reduction**: project the data on a smaller space,

$$\mathbf{x} \in \mathbb{R}^D \;\mapsto\; \mathbf{z} \in \mathbb{R}^d, \qquad d \ll D,$$

keeping their structure; then look at the points $\mathbf{z}_i$: do they form groups?



2

Principal Component Analysis

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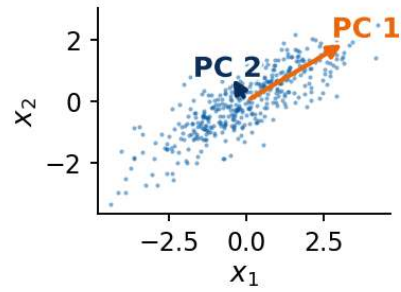
Principal Component Analysis: a Linear Projection

Principal component analysis (PCA): data $\mathbf{x}_1, \ldots, \mathbf{x}_n \in \mathbb{R}^D$, with mean $\bar{\mathbf{x}} = \frac{1}{n}\sum_{i=1}^n \mathbf{x}_i$ (for MNIST: the average image).

Covariance matrix of the features, $D \times D$:

$$C_{jl} = \frac{1}{n}\sum_{i=1}^{n}(x_{ij} - \bar{x}_j)(x_{il} - \bar{x}_l), \quad j, l = 1, \ldots, D$$

x_{ij}: feature j (e.g. pixel j) of sample i; C_{jj}: the variance of feature j; C_{jl}: how features j and l vary together over the data.



400 points in $\mathbb{R}^2$; PC 1 (first principal direction): 93.4% of the variance

K. Pearson, *On lines and planes of closest fit to systems of points in space*, Phil. Mag. **2**, 559 (1901), doi:10.1080/14786440109462720; H. Hotelling, *Analysis of a complex of statistical variables into principal components*, J. Educ. Psychol. **24**, 417 (1933), doi:10.1037/h0071325

Principal Directions Are Eigenvectors of the Covariance

PCA looks for the directions along which the data vary most:

- diagonalise C: $C\,\boldsymbol{\lambda}_l = \sigma_l^2\,\boldsymbol{\lambda}_l, \quad \sigma_1^2 \geq \sigma_2^2 \geq \cdots \geq \sigma_D^2 \geq 0, \quad l = 1, \ldots, D$
- $\boldsymbol{\lambda}_l$: the l-th **principal direction**, a unit eigenvector of C; the eigenvalue σ_l^2 is the variance of the data along $\boldsymbol{\lambda}_l$
- keep the k directions with the largest variances, the **principal components**; they keep the fraction $\sum_{l \leq k} \sigma_l^2 \Big/ \sum_{l \leq D} \sigma_l^2$ of the variance
- collect them as the columns of the $D \times k$ matrix $\Lambda_k = (\boldsymbol{\lambda}_1, \ldots, \boldsymbol{\lambda}_k)$

Features divided by their standard deviations first: C becomes the *correlation* matrix.

Each Sample Becomes a Point of the New Plane

Take the sample centred, $\mathbf{x}_i \to \mathbf{x}_i - \bar{\mathbf{x}}$. Its coordinate along the l-th principal direction is

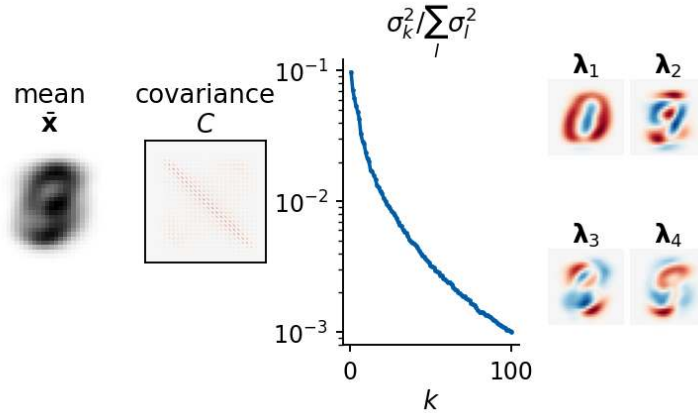
$$w_i^{(l)} = \boldsymbol{\lambda}_l^\top \mathbf{x}_i, \qquad l = 1, \ldots, k,$$

the length of the projection of $\mathbf{x}_i$ on $\boldsymbol{\lambda}_l$. The k coordinates form the **new representation** of $\mathbf{x}_i$:

$$\mathbf{w}_i = \left[w_i^{(1)}, \ldots, w_i^{(k)} \right]^\top = \Lambda_k^\top \mathbf{x}_i \in \mathbb{R}^k$$

- $k = 2$: $\mathbf{w}_i$ is the point plotted in the plane; MNIST: $\mathbf{x}_i \in \mathbb{R}^{784} \mapsto \mathbf{w}_i \in \mathbb{R}^2$
- the map is linear and fixed after fitting $(\bar{\mathbf{x}}, \Lambda_k)$: it applies to new samples as well

PCA on MNIST: the Ingredients

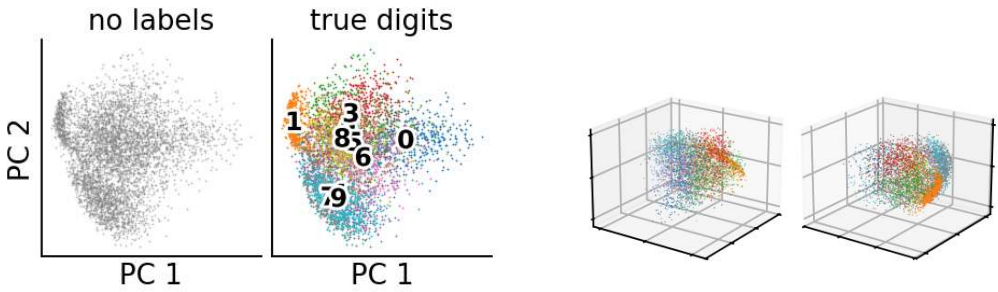


$\bar{\mathbf{x}}$, C, variance fraction $\sigma_k^2/\sum_l \sigma_l^2$, principal directions $\boldsymbol{\lambda}_l$

PCA on MNIST: from 784 Pixels to a Few Numbers

- from left to right, the figure shows the mean image $\bar{\mathbf{x}}$, the covariance C, the variance fraction $\sigma_k^2 / \sum_l \sigma_l^2$ of the k-th component against k, and the first four principal directions as 28×28 images, with red for positive and blue for negative entries
- 60 000 training images, each a vector of 784 pixels; C is 784×784: how two pixels vary together over the images
- the eigenvalues σ_l^2 fall fast: 2 components keep 17% of the variance, 10 keep 49%, 87 components keep 90%
- the principal directions $\boldsymbol{\lambda}_l$ are themselves images: patterns of pixels that vary together
- each digit becomes $\mathbf{w}_i = \Lambda_2^\top (\mathbf{x}_i - \bar{\mathbf{x}}) \in \mathbb{R}^2$, the two numbers plotted on the next frame

PCA on MNIST

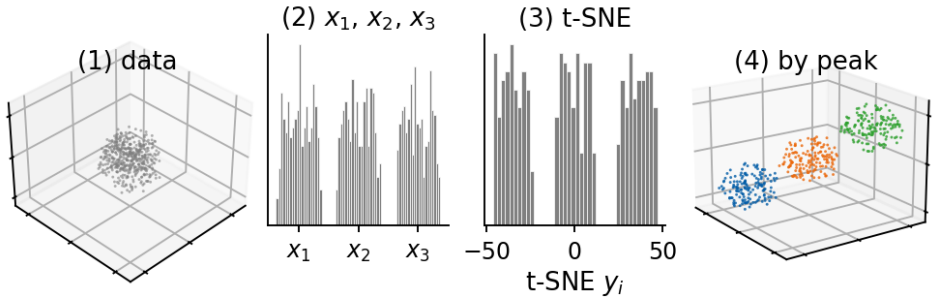


- 5000 test digits on 2 components (left): 17% of the variance; on 3 (right, two views, colours as in the middle panel): 23%; on 50: 82%
- is there clustering? some digits form separate groups (0, 1), the third direction separates a few more, but most overlap: the digits do not lie in a flat, low-dimensional subspace, and a linear projection is too crude

3
t-SNE

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t-SNE: from 3D to a Line



450 points in $\mathbb{R}^3$, given *without* labels: (1) the cloud seen along the diagonal; (2) histograms of the three coordinates x_1, x_2, x_3; (3) histogram of the single number y_i that t-SNE (t-distributed stochastic neighbour embedding) gives each point; (4) the points in 3D, coloured by their peak in (3)

t-SNE Finds Three Groups that No Coordinate Shows

- 450 points in $\mathbb{R}^3$, given *without* labels; seen along the diagonal they form one blob (1), and along every coordinate the values overlap (2): no single coordinate shows how many groups there are (in 3D one can still rotate the picture; in 100 dimensions there is no view to look from)
- t-SNE gives each point one number y_i, keeping close points close; on the line three separate peaks appear (3)
- colouring the points by their peak reveals three groups in 3D (4), exactly the three that generated the data
- what is kept: who is next to whom; what is not: large distances, so the gaps between the peaks are not the distances between the groups

t-SNE: Distances in the Original Space

A linear projection to 2D squeezes curved structure together. t-SNE asks instead: **neighbours in $\mathbb{R}^D$ stay neighbours in the 2D map**. The positions in the map are the parameters, found by gradient descent; there is no network.

$\mathbf{x}_i \in \mathbb{R}^D$, $i = 1, \ldots, n$: the original data instances (one MNIST image: $D = 784$). Each pair is reduced to one number, the squared distance

$$\|\mathbf{x}_i - \mathbf{x}_j\|^2 = \sum_{k=1}^{D} (x_{ik} - x_{jk})^2,$$

a sum over the D features (for images i and j: over their pixels k).

t-SNE: Similarities in the Original Space

For instance i, turn its distances to all other instances into weights with a Gaussian of width σ_i, and divide by their sum:

$$p_{j|i} = \frac{e^{-\|\mathbf{x}_i - \mathbf{x}_j\|^2/2\sigma_i^2}}{\sum_{l \neq i} e^{-\|\mathbf{x}_i - \mathbf{x}_l\|^2/2\sigma_i^2}}, \qquad p_{i|i} = 0$$

the probability that $\mathbf{x}_i$ would pick $\mathbf{x}_j$ as its neighbour, out of the $n-1$ other instances: a sum over the data set, not an integral over $\mathbb{R}^D$. σ_i is set by the **perplexity**, roughly the number of neighbours each instance keeps (here 30). Symmetrised over the pair: $p_{ij} = (p_{j|i} + p_{i|j})/2n$, $\sum_{i \neq j} p_{ij} = 1$.

$P = \{p_{ij}\}$

a probability distribution over all pairs of instances: how likely i and j are neighbours *in the original space*

t-SNE: Similarities in the Map

$\mathbf{y}_i \in \mathbb{R}^d$: the point that represents instance i in a **low-dimensional space whose dimension d we choose**; $d = 2$ to plot it. The $\mathbf{y}_i$ are the unknowns.

Similarity of two map points, with a Student-t:

$$q_{ij} = \frac{(1 + \|\mathbf{y}_i - \mathbf{y}_j\|^2)^{-1}}{\sum_{l \neq m}(1 + \|\mathbf{y}_l - \mathbf{y}_m\|^2)^{-1}}$$

with $\|\mathbf{y}_i - \mathbf{y}_j\|^2 = \sum_{a=1}^{d}(y_{ia} - y_{ja})^2$, a sum over the d map coordinates; the denominator sums over all pairs, so $\sum_{i \neq j} q_{ij} = 1$.

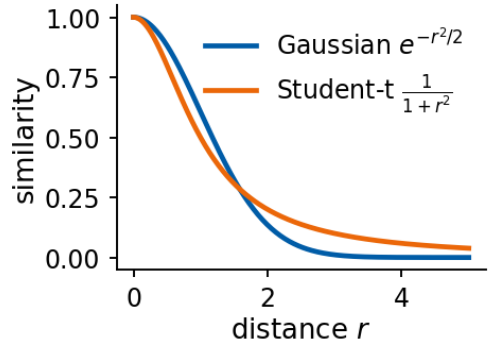
$Q = \{q_{ij}\}$

the same kind of distribution over pairs of instances, *in the map*

The Heavy Tail Opens Gaps between Groups

In 784 dimensions an instance can have many neighbours at similar distances; in 2D there is no room for all of them (the *crowding problem*).

The Student-t decays more slowly than the Gaussian, so moderately distant instances sit further apart in the map, and gaps open between the groups.



similarity against distance r: Gaussian $e^{-r^2/2}$ (original space), Student-t $1/(1+r^2)$ (map)

t-SNE: the Loss

Two distributions over pairs of instances: P from the original space, Q from the map. Move the map points $\mathbf{y}_i$ by gradient descent until Q matches P, i.e. minimise their Kullback–Leibler (KL) divergence:

$$\mathcal{L}(\mathbf{y}_1, \ldots, \mathbf{y}_n) = \mathrm{KL}(P \,\|\, Q) = \sum_{i \neq j} p_{ij} \log \frac{p_{ij}}{q_{ij}}$$

- neighbours placed far apart (p_{ij} large, q_{ij} small): large cost
- distant instances placed close (p_{ij} small): small cost
- so t-SNE keeps the **local** structure; distances between groups in the map carry little meaning

The result is a set of 2D points, one per instance, that we can plot and inspect.

L. van der Maaten, G. Hinton, *Visualizing Data using t-SNE*, J. Mach. Learn. Res. **9**, 2579 (2008), jmlr.org/papers/v9/vandermaaten08a

The t-SNE Map Points Are Free Parameters

Both methods give a **new representation** of each sample, in different ways:

PCA: a formula

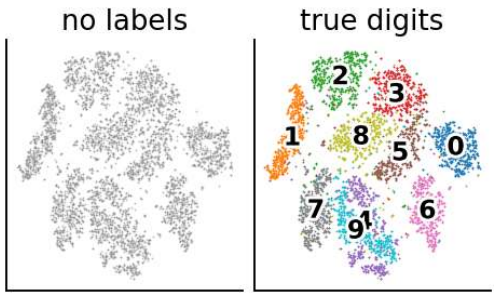
$\mathbf{x}_i \in \mathbb{R}^D \mapsto \mathbf{w}_i = \Lambda_k^\top \mathbf{x}_i \in \mathbb{R}^k$ (centred $\mathbf{x}_i$): a linear map, fixed after fitting; it applies to new samples

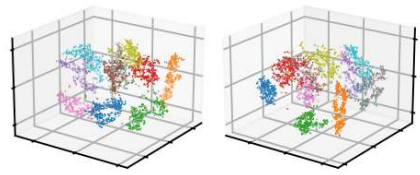
t-SNE: an optimisation

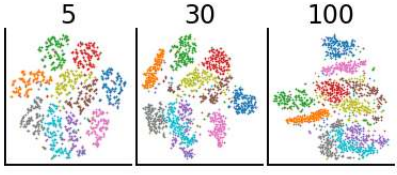
$\mathbf{x}_i \in \mathbb{R}^D \mapsto \mathbf{y}_i \in \mathbb{R}^2$, but $\mathbf{y}_i$ is not given by a formula: the $\mathbf{y}_i$ are free parameters, found by minimising $\mathrm{KL}(P \,\|\, Q)$ between the neighbour probabilities p_{ij} (Gaussian, input space) and q_{ij} (Student-t, map)

For t-SNE there is no map for new points, and only the neighbourhoods are meaningful.

t-SNE on MNIST



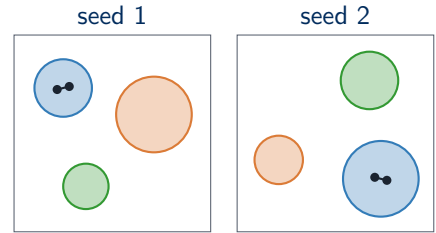




top: the 5000 digits in 2D (labels only colour the second panel) and in 3D (same colours); left: 2000 digits at perplexity 5, 30, 100. Ten groups appear without any label; small perplexity keeps only very close neighbours, large perplexity smooths the map

Only the Neighbourhoods in a t-SNE Map Are Reliable

- **meaningless:** the position of a cluster (a rotated, mirrored or shifted map has the same loss; another seed moves it)
- **meaningless:** the **separation** between clusters (distant pairs have $p_{ij} \approx 0$, so the loss hardly depends on it)
- **meaningless:** cluster **sizes** (σ_i adapts to the density) and larger distances **inside** a cluster
- **reliable:** which instances are close neighbours



Schematic of two t-SNE runs on the same data with different random seeds. The groups move, mirror and change size and separation, while the two neighbouring instances (black) stay together.

4

Clustering



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k-means Clustering

Goal: split the n data instances into k groups (clusters), such that instances in the same group are close to each other.
Each cluster S_c (the set of its instances) has a centre $\boldsymbol{\mu}_c$; k-means minimises the total squared distance of the instances to their centres:

$$\mathcal{L} = \sum_{c=1}^{k} \sum_{i \in S_c} \|\mathbf{x}_i - \boldsymbol{\mu}_c\|^2$$

The algorithm

start from k random centres, then alternate *assign* (each instance joins its nearest centre) and *update* (each centre moves to the mean of its instances), until no instance changes cluster

S. P. Lloyd, *Least squares quantization in PCM*, IEEE Trans. Inf. Theory **28**, 129 (1982), doi:10.1109/TIT.1982.1056489

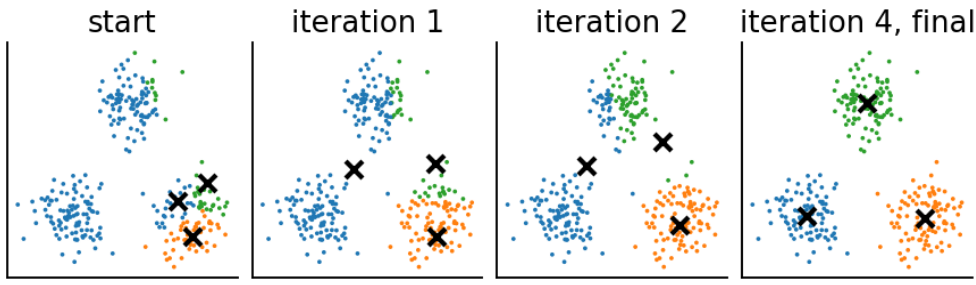
The k-means Algorithm Always Stops

The iterative algorithm:

1. choose k and place the k centres $\boldsymbol{\mu}_1, \ldots, \boldsymbol{\mu}_k$ at random (e.g. at k randomly chosen instances)
2. **assign:** each instance joins the cluster of its nearest centre, $S_c = \{\, i : \|\mathbf{x}_i - \boldsymbol{\mu}_c\| \leq \|\mathbf{x}_i - \boldsymbol{\mu}_{c'}\| \text{ for all } c' \,\}$
3. **update:** each centre moves to the mean of its instances, $\boldsymbol{\mu}_c = \frac{1}{|S_c|} \sum_{i \in S_c} \mathbf{x}_i$
4. repeat 2 and 3 until no instance changes cluster

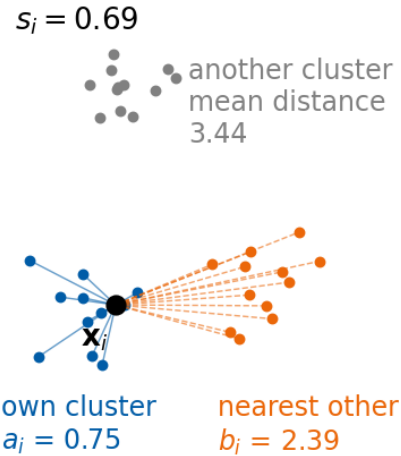
- each move can only lower $\mathcal{L}$, so the algorithm always stops
- it can stop in a poor solution: run it from several random starts and keep the lowest $\mathcal{L}$
- k is chosen by us

k-means, Iteration by Iteration



Three groups of points in 2D, the three centres (crosses) placed badly on purpose; colours show the assignment. The loss never increases, and after a few iterations nothing changes.

The Silhouette Score



In a good clustering each instance is **close to its own cluster** and **far from the other clusters**. The silhouette measures this for every instance, by comparing two mean distances.

For an instance $\mathbf{x}_i$ in cluster S_c:

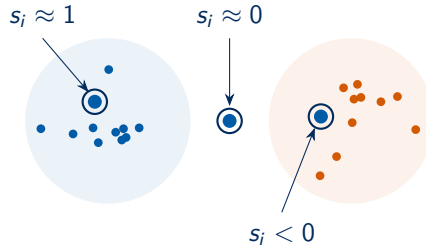
- a_i: mean distance from $\mathbf{x}_i$ to the other instances of S_c
- b_i: mean distance from $\mathbf{x}_i$ to the instances of the nearest other cluster

$$s_i = \frac{b_i - a_i}{\max(a_i, b_i)} \in [-1, 1]$$

The Mean Silhouette Chooses the Number of Clusters

$$s_i = \frac{b_i - a_i}{\max(a_i, b_i)} \in [-1, 1]$$

- $s_i \approx 1$: well inside its cluster, far from the others
- $s_i \approx 0$: on the border between two clusters
- $s_i < 0$: closer to another cluster, probably assigned wrongly

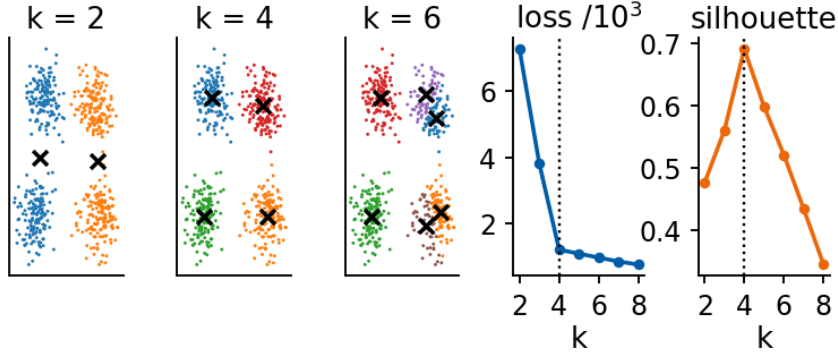


Schematic. The three circled instances belong to the blue cluster.

$\bar{s}(k) = \frac{1}{n}\sum_i s_i$ for the clustering with k clusters; choose the k with the largest $\bar{s}(k)$.

P. J. Rousseeuw, *Silhouettes: A graphical aid to the interpretation and validation of cluster analysis*, J. Comput. Appl. Math. **20**, 53 (1987), doi:10.1016/0377-0427(87)90125-7

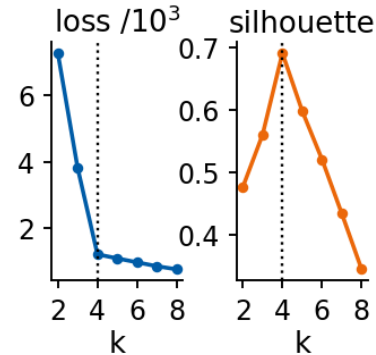
Choosing the Number of Clusters



k-means needs k from us. For every $k = 2, 3, \ldots, 8$: run k-means (10 random starts, keep the lowest loss), record the loss $\mathcal{L}(k)$ and the mean silhouette score $\bar{s}(k)$, and plot both against k. Four groups of points, so the right answer is known.

Elbow and Silhouette Both Point to Four Clusters

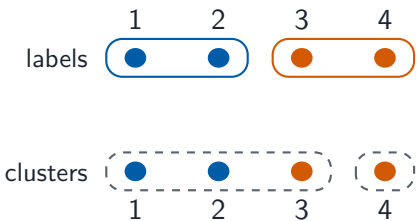
- **elbow:** $\mathcal{L}(k)$ always decreases with k (more centres, shorter distances); it decreases fast up to the right k and slowly after it: the bend is at $k = 4$
- **silhouette:** $\bar{s}(k)$ is highest at $k = 4$, with 0.69 against 0.56 at $k = 3$ and 0.60 at $k = 5$
- too small k merges groups and too large k splits them, as the panels for $k = 2$ and $k = 6$ show



loss $\mathcal{L}(k)$ and mean silhouette $\bar{s}(k)$ for the data with four groups

Rand Index

How well does a clustering recover known classes? Compare two partitions of the same n instances, the true labels and the clusters, *pair by pair*; the names of the groups do not matter.

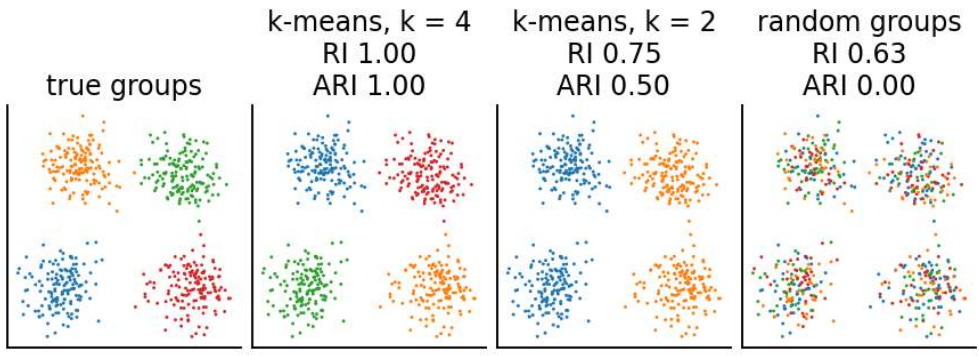


pair	labels	clusters	agree
1–2	same	same	✓
1–3	different	same	×
1–4	different	different	✓
2–3	different	same	×
2–4	different	different	✓
3–4	same	different	×

A pair **agrees** if both partitions put it in the same group, or both in different groups. The **Rand index** is the fraction of agreeing pairs, out of all $n(n-1)/2$:

$$\mathrm{RI} = \frac{\text{agreeing pairs}}{n(n-1)/2} = \frac{3}{6} \quad \text{here;} \qquad \mathrm{RI} = 1 \text{ for identical partitions}$$

Random Groups Already Reach a High Rand Index



Random groups already reach a high RI (0.63 here): most pairs are in different groups in both partitions.

Adjusted Rand Index

- the adjusted Rand index corrects for chance:

$$\mathrm{ARI} = \frac{\mathrm{RI} - \mathbb{E}[\mathrm{RI}]}{1 - \mathbb{E}[\mathrm{RI}]},$$

  with $\mathbb{E}[\mathrm{RI}]$ the mean RI when the instances are reassigned at random, keeping the group sizes

- **ARI = 1**: the same partition; **ARI $\approx$ 0**: no better than chance; it can be negative

partition	RI	ARI
k-means, $k = 4$	0.998	0.996
k-means, $k = 2$	0.748	0.495
random groups	0.626	0.001

the data with four groups and the partitions of the Rand-index figure

L. Hubert and P. Arabie, *Comparing partitions*, J. Classification **2**, 193 (1985), doi:10.1007/BF01908075

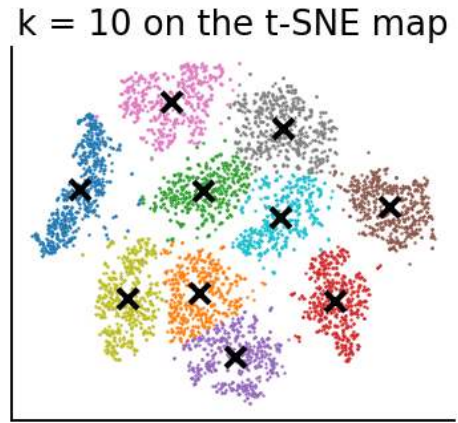
k-means on MNIST

Split the samples into k groups $S_1, \ldots, S_k$ with centres $\boldsymbol{\mu}_1, \ldots, \boldsymbol{\mu}_k$, minimising

$$\mathcal{L} = \sum_{c=1}^{k} \sum_{i \in S_c} \|\mathbf{x}_i - \boldsymbol{\mu}_c\|^2$$

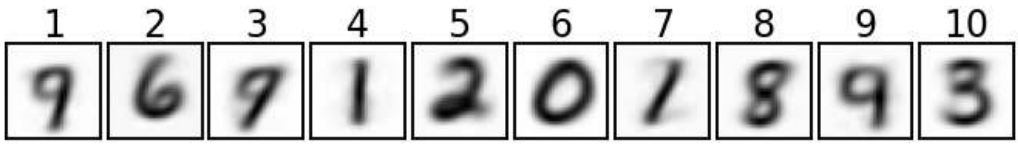
Repeat until nothing changes:

- assign every sample to its nearest centre
- move every centre to the mean of its samples



Ten Centres and the Agreement with the Digits

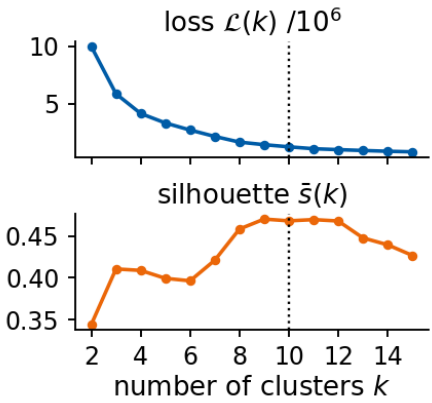
$k = 10$, the number of digits. Agreement with the true digits, adjusted Rand index: 0.38 on 50 PCA components, 0.76 on the t-SNE map.



the ten centres $\boldsymbol{\mu}_1, \ldots, \boldsymbol{\mu}_{10}$ found on 50 PCA components, shown as images

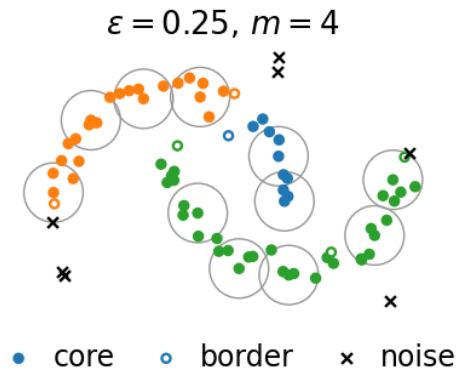
S. P. Lloyd, *Least squares quantization in PCM*, IEEE Trans. Inf. Theory **28**, 129 (1982), doi:10.1109/TIT.1982.1056489

Choosing k on MNIST



- the same two guides on the t-SNE map of the 5000 digits: the loss $\mathcal{L}(k)$ and the mean silhouette $\bar{s}(k) = \frac{1}{n}\sum_i \frac{b_i - a_i}{\max(a_i, b_i)}$ (a_i: mean distance to its own cluster, b_i: to the nearest other cluster)
- dotted line: $k = 10$, the number of digits; drawn only for comparison, the guides do not use the labels
- no sharp elbow; the silhouette is highest at $k = 9$, and almost equal at $k = 10$ and 11: a flat maximum next to the 10 digits (at $k = 10$ the 4s and 9s are mixed across two clusters)
- on real data the guides suggest a range of k, not one answer: the final choice uses what we know about the data

DBSCAN: Clustering by Density



DBSCAN (density-based spatial clustering of applications with noise) has two parameters: a radius ε and a minimum number of points m.

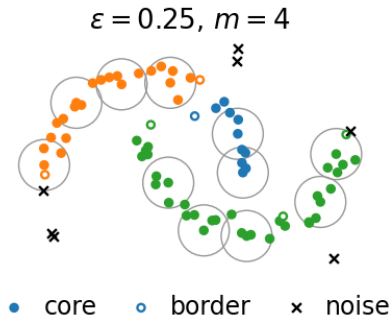
- **core point:** at least m points within distance ε (grey circles)
- **border point:** not core, but within ε of a core point
- **noise:** all other points (crosses)

No number of clusters is needed, and clusters can have any shape.

M. Ester, H.-P. Kriegel, J. Sander, X. Xu, *A Density-Based Algorithm for Discovering Clusters in Large Spatial Databases with Noise*, KDD 1996, AAAI proceedings

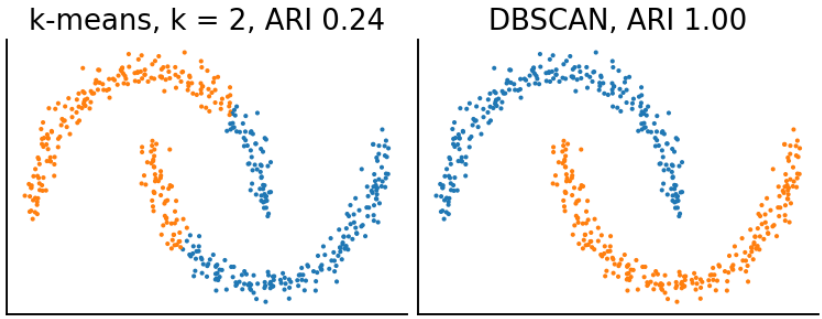
The DBSCAN Algorithm

1. count the neighbours within ε of every point (itself included) and mark the core points
2. take a core point not yet in a cluster and start a new cluster with it
3. add every point within ε of a core point of the cluster; repeat until no point is added: the cluster grows through the dense region and stops at its border points
4. repeat 2 and 3 until every core point is in a cluster; the points never reached are noise



With $\varepsilon = 0.25$ and $m = 4$ the algorithm finds 3 clusters, with 65 core, 6 border and 7 noise points.

DBSCAN Finds Clusters of Any Shape

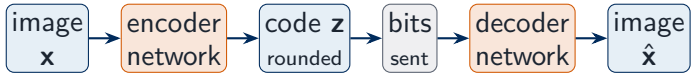


- two half-moons, 500 points
- k-means cuts the space into convex cells, so it finds round clusters: with $k = 2$ it gives ARI 0.24
- DBSCAN follows the dense regions: ARI 1.00

5
Autoencoders

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Compression with a Neural Network



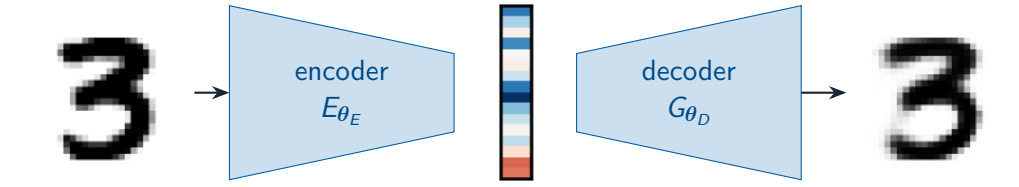
learned model of $p(\mathbf{z})$: frequent codes get short bit strings

- an autoencoder whose code is **rounded to integers** and written as bits; the decoder rebuilds an approximation $\hat{\mathbf{x}}$
- trained end to end on a **rate–distortion** loss $\mathcal{L} = R + \kappa D$, where the weight κ sets the trade-off, with R the expected number of bits (from a learned probability model of the code) and $D = \|\mathbf{x} - \hat{\mathbf{x}}\|^2$
- Ballé et al. formulate this as a variational autoencoder, and learned image coding has been standardised as JPEG AI; the codecs common on phones today (JPEG, HEIC) still use hand-designed transforms

J. Ballé, V. Laparra, E. P. Simoncelli, arXiv:1611.01704; J. Ballé et al., arXiv:1802.01436; E. Alshina, J. Ascenso, T. Ebrahimi, IEEE MultiMedia **31** (2024), doi:10.1109/MMUL.2024.3485255

Autoencoder

PCA compresses with a straight (linear) projection: MNIST needs 87 components for 90% of the variance. Can a network learn a better, non-linear compression? *If a network can rebuild an image from only d numbers, those numbers must contain what is essential about it.*



G. E. Hinton, R. R. Salakhutdinov, *Reducing the Dimensionality of Data with Neural Networks*, Science **313**, 504 (2006), doi:10.1126/science.1127647

Encoder, Decoder and the Latent Space

- **encoder** E_{θ_E}: a neural network with D inputs (the size of the data, 784 pixels) and d outputs (the latent-space size, chosen by us, here 16); its own parameters (weights and biases) $\boldsymbol{\theta}_E$; $\mathbf{z} = E_{\theta_E}(\mathbf{x}) \in \mathbb{R}^d$
- **decoder** G_{θ_D}: a second neural network with d inputs and D outputs, with its own parameters $\boldsymbol{\theta}_D$; $\hat{\mathbf{x}} = G_{\theta_D}(\mathbf{z}) \in \mathbb{R}^D$
- both are trained together on the images alone, with no labels: every input is its own target
- the bottleneck $d \ll 784$ forces the encoder to extract the crucial features of the data, those the decoder needs to rebuild the image; they are encoded in $\mathbf{z}$, a learned, low-dimensional **latent-space representation**
- a linear autoencoder finds the same subspace as PCA; non-linear layers can follow curved structure

The Autoencoder Loss

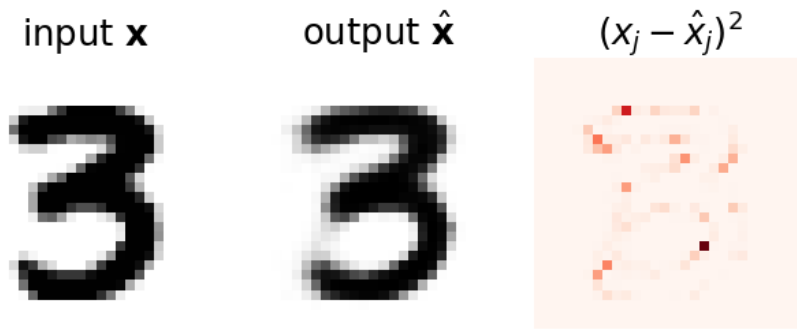
A **loss function** is one number that measures how badly the network does on the training data, for given values of its parameters; training means finding the parameters that make it smallest. Here it compares each input with its reconstruction $\hat{\mathbf{x}}_i = G_{\theta_D}(E_{\theta_E}(\mathbf{x}_i))$ **pixel by pixel**:

$$\mathcal{L}(\boldsymbol{\theta}_E, \boldsymbol{\theta}_D) = \frac{1}{n}\sum_{i=1}^{n}\frac{1}{D}\sum_{j=1}^{D}\left(x_{ij} - \hat{x}_{ij}\right)^2$$

i: image, j: pixel.

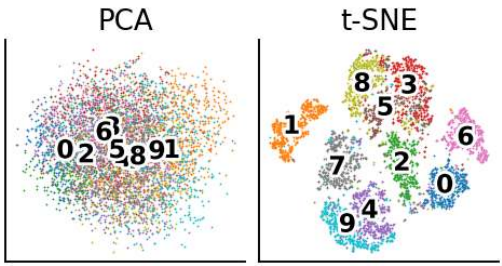
- $\mathcal{L} = 0$ only if every pixel is rebuilt exactly
- no labels: the target is the input itself
- $\boldsymbol{\theta}_E$ and $\boldsymbol{\theta}_D$ are updated together by gradient descent and backpropagation, as for any network; for MNIST: dense layers $784 \to 256 \to 64 \to d$ and back

Reconstruction Error of One Digit



The right panel is the squared difference $(x_{ij} - \hat{x}_{ij})^2$ for one digit; $\mathcal{L}$ is its average over all pixels and all images.

Compression and the Latent Space



16 latent numbers of each of 5000 test digits, in 2D; colours: true digits (plotting only)

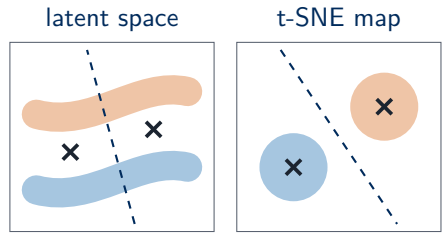
- **compression:** a 784-pixel image becomes $d = 16$ numbers $\mathbf{z} = E_{\theta_E}(\mathbf{x})$ (49 times fewer), from which the decoder rebuilds it
- the loss does not ask for it, but similar images end up with nearby $\mathbf{z}$: the decoder must rebuild them alike
- **PCA of z:** a linear projection onto the two directions of largest variance; the digits differ along many of the 16 directions, so in any two of them they overlap
- **t-SNE of z:** keeps the neighbours of each point in 16D and shows the groups

The Latent Space Keeps the Digits, Not as Round Groups

k-means ($k = 10$) against the true digits, adjusted Rand index (ARI):

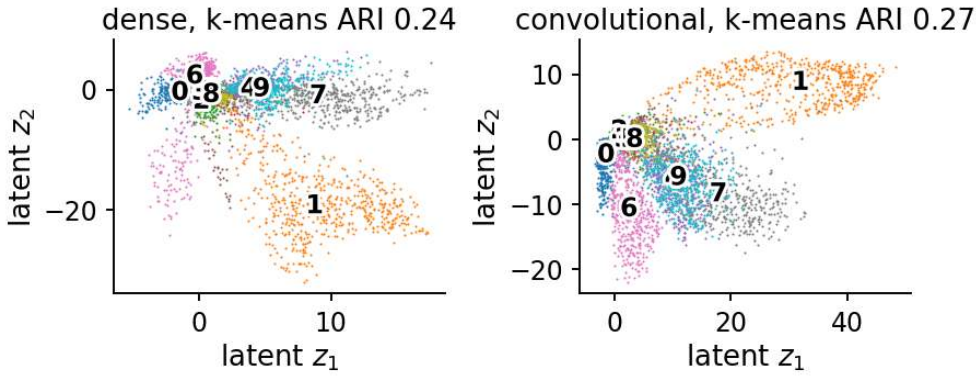
	directly	t-SNE map
pixels (50 PCA comp.)	0.38	0.76
16 latent numbers	0.28	0.70

- the digits are kept in the latent space, but not as the round groups that k-means looks for
- t-SNE only *draws* the latent space; the latent space itself takes *new* images (encoder) and gives back images (decoder)



Schematic with two groups. k-means splits the space by a straight boundary (dashed) between its centres (crosses). It cuts elongated groups across and separates compact ones.

Autoencoder on MNIST: Dense and Convolutional



Autoencoders with only $d = 2$ latent numbers, so the latent space can be drawn directly, without PCA or t-SNE; colours: true digits.

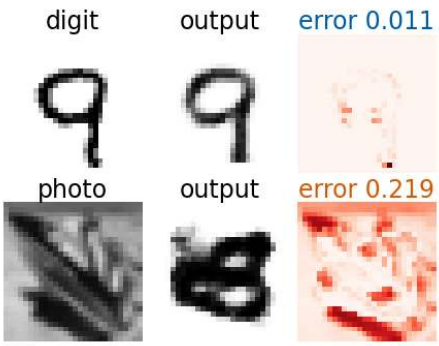
Dense and Convolutional Encoders with Two Latent Numbers

- **dense** (left): sees an image as a flat list of 784 pixels; **convolutional** (right): two convolutions with stride 2 ($28 \to 14 \to 7$ pixels, 16 and 32 channels), as in a CNN
- with the same two numbers, the convolutional encoder separates the digits better: it compares neighbouring pixels, as a CNN does

5000 test digits, $d = 2$	dense	convolutional
reconstruction error (mean squared error per pixel)	0.039	0.045
ARI of k-means ($k = 10$) in the latent space	0.24	0.27

The convolutional encoder reconstructs slightly worse and still groups the digits better.

Anomaly Detection with an Autoencoder



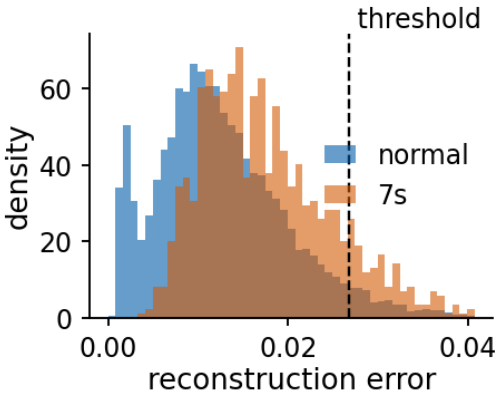
autoencoder trained on all MNIST digits; a typical digit and a typical photograph (median errors)

The task: find samples that are unlike the rest, when we have no examples of them.

- a well-trained autoencoder has a **small loss** on data like its training data: it rebuilds them well
- very odd data are rebuilt badly: the reconstruction error $\|\mathbf{x} - G_{\theta_D}(E_{\theta_E}(\mathbf{x}))\|^2$ is **large**
- a large error means the sample is *surprising* for the model: an outlier, an **anomaly**

Decision rule: choose a threshold on the error, e.g. so that only 5% of normal samples exceed it (5% false alarms), and flag every sample above it.

A Subtle Anomaly: the Digit 7



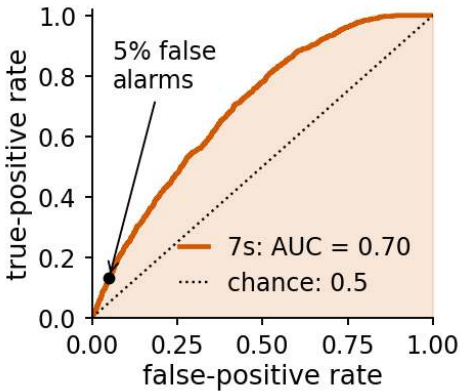
- **experiment:** train the autoencoder on all digits *except the 7s*, then compute the error of every test digit
- **histogram:** horizontal axis = reconstruction error; blue: the digits it was trained on, orange: the 7s, never seen
- the two populations overlap: the threshold (dashed, 5% false alarms) flags only 13% of the 7s; AUC 0.70
- a 7 shares strokes with 1 and 9, so many 7s are rebuilt well

The Largest Errors Mix 7s and Badly Written Digits



The 16 test digits with the largest error, labelled by their true digit: some are 7s (orange labels), others are badly written digits of the trained classes.

ROC Curve and AUC



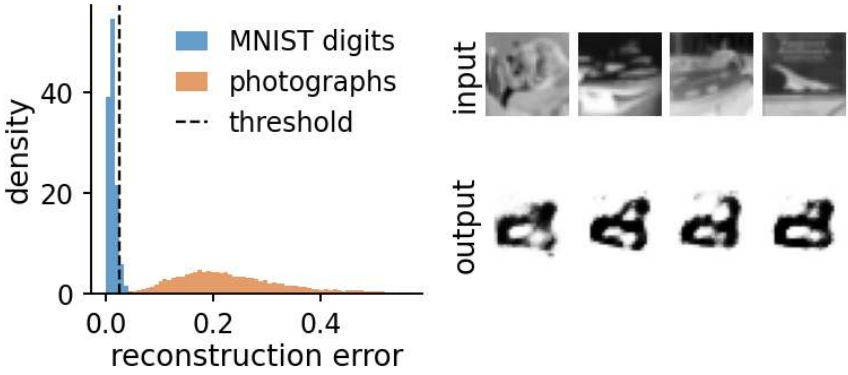
Each threshold on the anomaly score gives one decision rule. For each:

- **true-positive rate** (TPR): fraction of anomalies above the threshold, i.e. found
- **false-positive rate** (FPR): fraction of normal samples above it, i.e. false alarms

Lowering the threshold traces the **ROC curve** (receiver operating characteristic) from $(0, 0)$ to $(1, 1)$. Chance lies on the diagonal; a perfect score goes straight up to TPR $= 1$.

AUC: the area under the curve, 0.5 for chance, 1 for a perfect separation; equal to the probability that a random anomaly has a larger score than a random normal sample.

A Clear Anomaly: Photographs



Experiment: the autoencoder trained on all MNIST digits receives CIFAR-10 photographs (aeroplanes, cars, animals, ships, trucks), converted to 28×28 grayscale.

Photographs Are Rebuilt as Blurred Digits

- the errors of digits and photographs barely overlap
- the reconstructions look like blurred digits: the decoder can only produce what it learned

anomaly	7s left out of training	photographs
reconstruction error, normal digits	0.013 (mean)	0.011 (median)
reconstruction error, anomalies	0.018 (mean)	0.219 (median)
flagged at 5% false alarms	13%	99.8%
AUC	0.70	0.9997

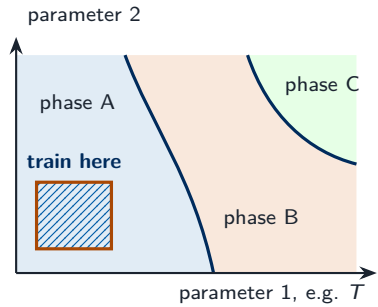
An anomaly far from the training data is easy to detect; one that resembles it is hard.

6

Anomaly Detection for Studies of Phase Diagrams

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Anomaly Detection on a Phase Diagram



schematic; thick lines: phase boundaries

- **train** the autoencoder only on a small, well-understood region of one phase (hatched)
- **scan:** compute the reconstruction error of samples everywhere in the diagram
- **plot** the error as a map: low inside the trained phase, high elsewhere
- **phase boundaries**: where the error changes fastest; no labels, no order parameter needed

Kottmann et al. mapped the phase diagram of the one-dimensional extended Bose–Hubbard model this way. K. Kottmann, P. Huembeli, M. Lewenstein, A. Acín, *Unsupervised Phase Discovery with Deep Anomaly Detection*, Phys. Rev. Lett. **125**, 170603 (2020), doi:10.1103/PhysRevLett.125.170603

Model by Model

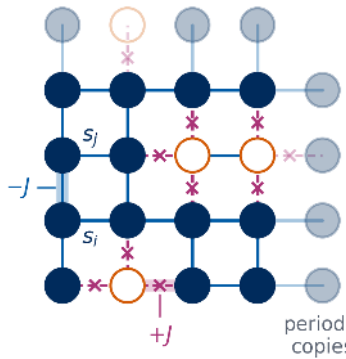
The same protocol for every model: **physics first** (Hamiltonian, phases, the observables physicists constructed), then the **unsupervised methods** (PCA, autoencoder anomaly detection), then a comparison with what is known.

	model	what makes it interesting
1	2D Ising	one phase transition, exactly known; the reference
2	J_1–J_2 Ising	competing couplings; the magnetisation misses a whole phase
3	2D XY	continuous spins; no local order parameter, a topological (vortex) transition

Model 1

The 2D Ising Model

The 2D Ising Model and Its Hamiltonian



$$H(\mathbf{s}) = -J \sum_{\langle i,j \rangle} s_i s_j$$

$J > 0$, $\langle i, j \rangle$: nearest neighbours
periodic boundaries (a torus)

$s_i = +1$

$s_i = -1$

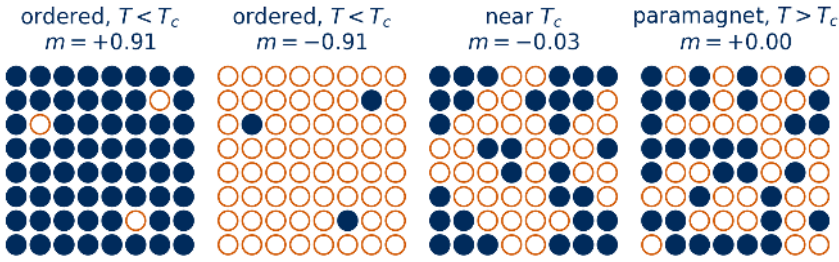
satisfied bond: $-J$

broken bond: $+J$

$p(\mathbf{s}) = e^{-\beta H(\mathbf{s})}/Z$, $\beta = 1/T$ $(k_B = J = 1)$; symmetry $H(-\mathbf{s}) = H(\mathbf{s})$: flipping every spin costs nothing

Lecture 4: a CNN with labels found T_c. Today: the same data, no labels.

Ordered and Disordered Phases



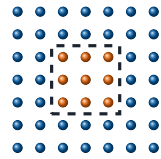
order parameter: magnetisation per spin $m = \frac{1}{N} \sum_i s_i$

	ordered, $T < T_c$	**disordered**, $T > T_c$
spins	mostly parallel, all up or all down	random, small domains
m	$\pm m_0(T) \neq 0$: symmetry $\mathbf{s} \to -\mathbf{s}$ broken	$m \to 0$ for $L \to \infty$

Domain-Wall Energy against Entropy

Equilibrium minimises $F = E - TS$, so order survives only if a large flipped domain raises F.

- In 1D a wall is one broken bond (energy $2J$) at any of $N - 1$ places. It changes F by $2J - T \ln(N - 1)$, negative for long chains at every $T > 0$.
- In 2D a closed wall of length ℓ costs $2J\ell$ and has at most about 3^ℓ shapes. It raises F by about $\ell\,(2J - T \ln 3)$, so order survives below $2J/\ln 3 = 1.82\,J$ (Peierls). The exact T_c is $2.269\,J$.
- The wall free energy per unit length, $\sigma(T) = 2J + T \ln \tanh(J/T)$ (Onsager), vanishes at T_c, and **domains of all sizes appear**.



wall of length $\ell = 12$

R. Peierls, Proc. Camb. Phil. Soc. **32**, 477 (1936), doi:10.1017/S0305004100019174; L. Onsager, Phys. Rev. **65**, 117 (1944), doi:10.1103/PhysRev.65.117

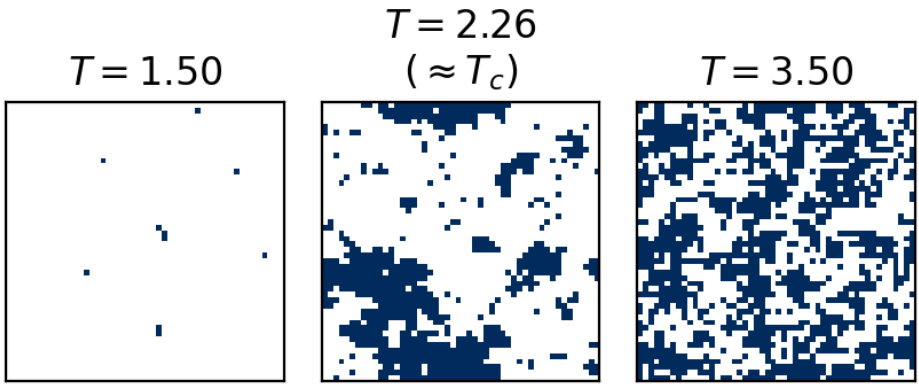
A Continuous Transition at $T_c = 2.269$

$T_c = 2/\ln(1+\sqrt{2}) = 2.269$ (Onsager); reduced temperature $t = (T - T_c)/T_c$

- **continuous** (second order): m vanishes without a jump, $m \sim (-t)^{\beta_M}$, $\beta_M = 1/8$; at a **first-order** transition the order parameter jumps
- correlation length $\xi \sim |t|^{-\nu}$, $\nu = 1$; susceptibility $\chi \sim |t|^{-\gamma}$, $\gamma = 7/4$; specific heat $c \sim -\ln|t|$
- the exponents are shared by all models with the same symmetry and dimension: the **2D Ising universality class**

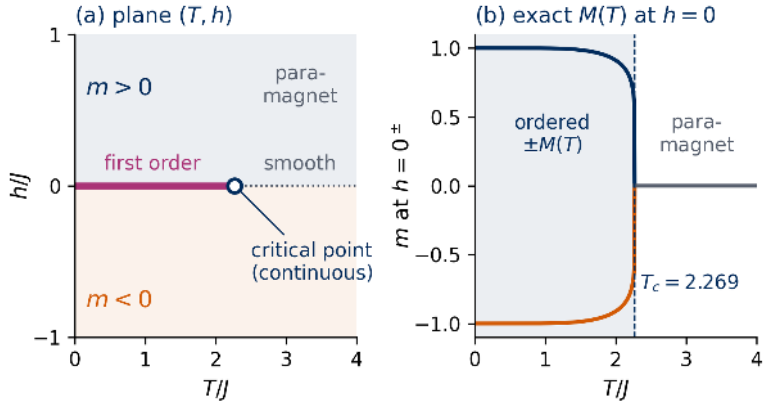
L. Onsager, Phys. Rev. **65**, 117 (1944), doi:10.1103/PhysRev.65.117; C. N. Yang, Phys. Rev. **85**, 808 (1952), doi:10.1103/PhysRev.85.808

Monte Carlo Snapshots across T_c



one Monte Carlo configuration per temperature: ordered below T_c, domains of all sizes near T_c, random above

Phase Diagram in the (T, h) Plane



field h: $H = -J\sum_{\langle i,j\rangle} s_i s_j - h\sum_i s_i$; $h = 0$, $T < T_c$: **first-order** line, m jumps from $-m_0$ to $+m_0$; it ends at the **critical point** $(T_c, 0)$; no transition for $T > T_c$. Our data: $h = 0$, a scan in T through the critical point.

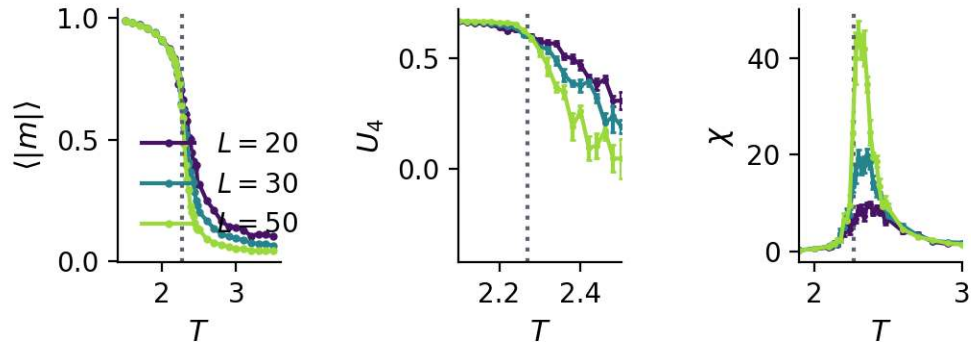
Observables: Definition and Estimator from Samples

$\langle O\rangle = \frac{1}{Z}\sum_{\mathbf{s}} O(\mathbf{s})\, e^{-\beta H(\mathbf{s})}$; Monte Carlo gives n samples $\mathbf{s}^{(k)} \sim p$, $e_k = H(\mathbf{s}^{(k)})/N$, $m_k = \frac{1}{N}\sum_i s_i^{(k)}$, $\overline{x} = \frac{1}{n}\sum_k x_k$

	definition through Z	estimator	measures
energy	$\langle e\rangle = -\frac{1}{N}\partial_\beta \ln Z$	$\overline{e}$	bonds
magnetisation	$\langle \lvert m\rvert\rangle = \frac{1}{Z}\sum_{\mathbf{s}} \lvert m(\mathbf{s})\rvert\, e^{-\beta H}$	$\overline{\lvert m\rvert}$	order
specific heat	$c = \frac{\beta^2}{N}\partial_\beta^2 \ln Z = N\beta^2(\langle e^2\rangle - \langle e\rangle^2)$	$N\beta^2(\overline{e^2} - \overline{e}^2)$	energy fluctuations
susceptibility	$\chi = N\beta(\langle m^2\rangle - \langle \lvert m\rvert\rangle^2)$	$N\beta(\overline{m^2} - \overline{\lvert m\rvert}^2)$	m fluctuations
Binder cumulant	$U_4 = 1 - \langle m^4\rangle/3\langle m^2\rangle^2$	$1 - \overline{m^4}/3\,\overline{m^2}^2$	shape of $P(m)$

$\langle m\rangle = 0$ on a finite lattice ($\mathbf{s} \to -\mathbf{s}$), hence $|m|$; χ with $\langle|m|\rangle$: the finite-lattice form, not $\partial_h \ln Z$; variances divide by n

The Reference: T_c from Finite-Size Scaling

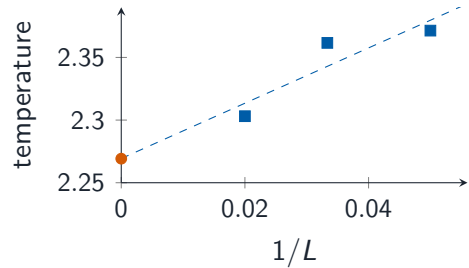


$m = \frac{1}{N}\sum_i s_i, \quad U_4 = 1 - \frac{\langle m^4\rangle}{3\langle m^2\rangle^2}, \quad \chi = \frac{N}{T}(\langle m^2\rangle - \langle|m|\rangle^2);$ lattices $L \times L$, $L = 20, 30, 50$

Binder Crossings and Susceptibility Peaks Give $T_c = 2.269$

We locate T_c from the magnetisation $\langle |m| \rangle$, the Binder cumulant U_4 and the susceptibility χ as functions of T.

- U_4 curves for different L **cross at** T_c: 2.250 ($L = 20/30$), 2.278 ($L = 30/50$)
- the peak of χ moves as $T_c(L) = T_c + a/L$: 2.371, 2.362, 2.303; fit: $T_c = 2.269$, exact 2.2692



squares: peaks of χ for $L = 50, 30, 20$; dashed: linear fit in $1/L$; dot: exact T_c

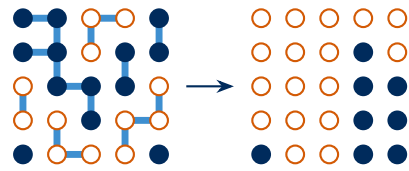
K. Binder, Z. Phys. B **43**, 119 (1981), doi:10.1007/BF01293604; M. E. J. Newman, G. T. Barkema, *Monte Carlo Methods in Statistical Physics*, Oxford University Press (1999)

Sampling near T_c: Flip Whole Clusters

Near T_c single-spin Metropolis flips are rejected inside large domains: successive samples are almost copies (*critical slowing down*).

Swendsen–Wang update:

1. bond each pair of parallel neighbours with probability $p = 1 - e^{-2\beta J}$
2. find the clusters connected by bonds
3. flip every cluster with probability $1/2$



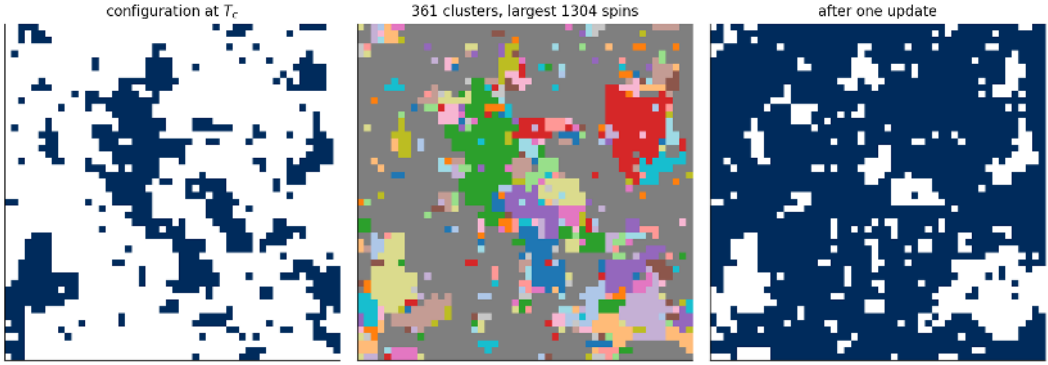
exact: $e^{\beta J s_i s_j} = e^{\beta J}\left[(1-p) + p\,\delta_{s_i s_j}\right]$

$L = 50$ at T_c: autocorrelation time 26.4 Metropolis sweeps vs 3.4 cluster updates

Filled circles $s_i = +1$, open circles $s_i = -1$. Bonds (light blue) join only parallel neighbours.

R. H. Swendsen, J.-S. Wang, Phys. Rev. Lett. **58**, 86 (1987), doi:10.1103/PhysRevLett.58.86

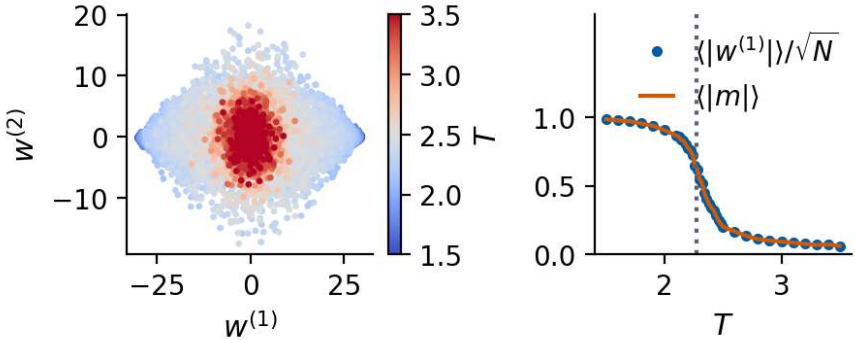
One Swendsen–Wang Update at T_c



left: a configuration at T_c; middle: the clusters connected by bonds, one colour per cluster; right: the configuration after every cluster was flipped with probability $1/2$. On this 50×50 lattice one update forms 361 clusters, and the largest contains 1304 of the 2500 spins.

PCA Finds the Order Parameter

9472 configurations (30×30, 37 temperatures), each a vector $\mathbf{s} \in \{-1, +1\}^{900}$; PCA receives only the spins, no temperature, no phase.



With the projections $w^{(l)} = \boldsymbol{\lambda}_l^\top(\mathbf{s} - \bar{\mathbf{s}})$, the left panel shows $(w^{(1)}, w^{(2)})$ of every configuration, coloured by T, and the right panel compares $\langle|w^{(1)}|\rangle/\sqrt{N}$ with $\langle|m|\rangle$.

Below T_c the Configurations Form Two Groups

The projections $w^{(l)}$ of the 9472 configurations on the principal directions $\boldsymbol{\lambda}_l$ arrange themselves by temperature.

- low T: two groups at $w^{(1)} \approx \pm 30 = \pm\sqrt{N}$, all spins up or all down; high T: one group around 0
- $\langle |w^{(1)}| \rangle / \sqrt{N}$ lies on the magnetisation $\langle |m| \rangle$ (correlation 0.99996 between $|w^{(1)}|$ and $|m|$); $\boldsymbol{\lambda}_1$ carries 37% of the variance
- the Binder crossings computed from $w^{(1)}$ agree with those from m

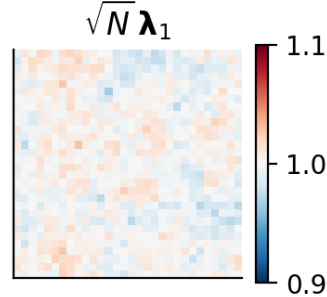
Binder crossing	from m	from $w^{(1)}$
$L = 20/30$	2.2495	2.2495
$L = 30/50$	2.2784	2.2773

L. Wang, Phys. Rev. B **94**, 195105 (2016), doi:10.1103/PhysRevB.94.195105; S. J. Wetzel, Phys. Rev. E **96**, 022140 (2017), doi:10.1103/PhysRevE.96.022140

The First Principal Component Is the Magnetisation

- below T_c all pairs of spins are correlated, $\langle s_i s_j \rangle \approx M^2$: the covariance matrix has a large uniform block, and its leading eigenvector is $\boldsymbol{\lambda}_1 \approx (1, \ldots, 1)/\sqrt{N}$
- projecting on it:

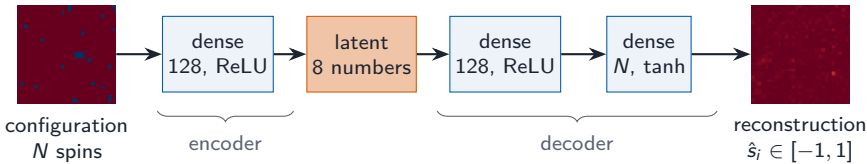
$$w^{(1)} = \boldsymbol{\lambda}_1^{\top}(\mathbf{s} - \bar{\mathbf{s}}) \approx \frac{1}{\sqrt{N}} \sum_{i=1}^{N} s_i = \sqrt{N}\, m$$



- measured: the entries of $\sqrt{N}\,\boldsymbol{\lambda}_1$ are 1 with a spread of 1%; the correlation of $|w^{(1)}|$ with $|m|$ is 0.99996

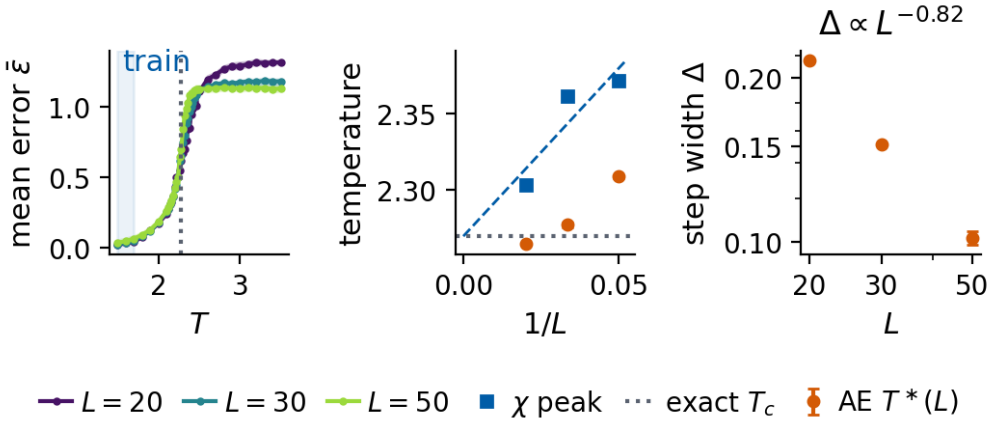
The Ising order parameter is linear in the spins, and a linear method finds linear structure.

Autoencoder on Spin Configurations: the Setup



- **input:** one configuration $\mathbf{s} \in \{-1, +1\}^N$, $N = L^2$, for $L = 20, 30, 50$; the same architecture for every L
- **training data:** only the 768 configurations with $T \leq 1.7$, deep in the ordered phase (Yang: $M(1.7) = 0.970$); no temperatures and no labels are given to the network
- **loss:** $\frac{1}{N}\|\mathbf{s} - \hat{\mathbf{s}}\|^2$, Adam, 200 epochs, three random seeds per L
- **measurement:** the mean reconstruction error at each of the 37 temperatures

The Autoencoder Signal Sharpens with the System Size



reconstruction error of one configuration $\varepsilon(\mathbf{s}) = \frac{1}{N}\sum_i (s_i - \hat{s}_i)^2$; $\bar{\varepsilon}(T) = \langle \varepsilon \rangle_T$, its average at T; fit $\bar{\varepsilon}(T) = a + b \tanh\big((T - T^*)/\Delta\big)$: centre T^*, width Δ

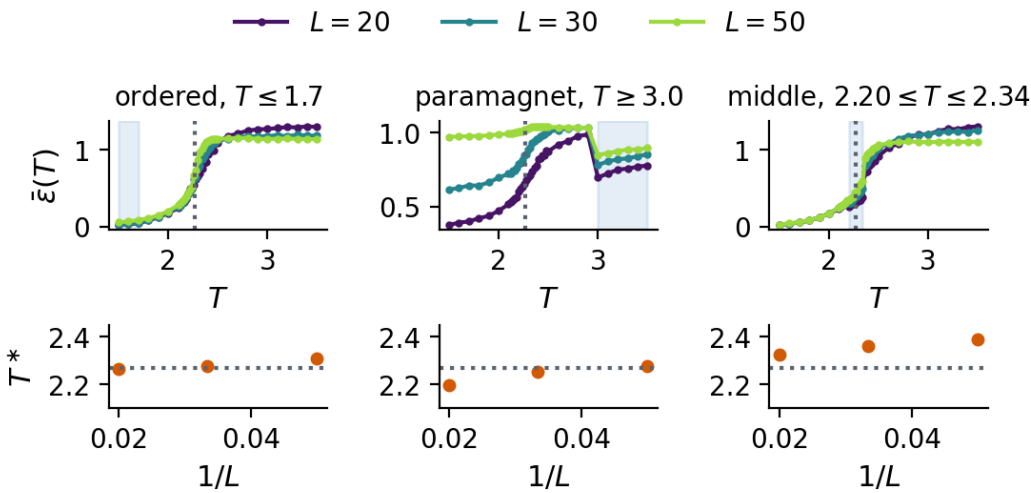
The Step Centre Approaches T_c and the Step Narrows

We train the autoencoder on $T \leq 1.7$ and fit a step to $\bar{\varepsilon}(T)$. The table lists its centre T^* and width Δ, each the mean over three seeds, next to the exact $T_c = 2/\ln(1+\sqrt{2}) = 2.269$.

L	T^*	$T^* - T_c$	Δ
20	2.308	$+0.039$	0.215
30	2.277	$+0.008$	0.151
50	2.264	-0.005	0.102

- $T^* \to T_c$ as L grows; three seeds agree to ± 0.002
- $\Delta \propto L^{-0.82}$; the finite-size rounding of the transition scales as $L^{-1/\nu} = L^{-1}$ (three sizes only: indicative)
- the autoencoder is not a thermodynamic observable: no derived finite-size law, so we do not extrapolate T^*

The Training Region Decides the Signal



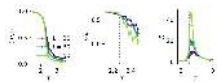
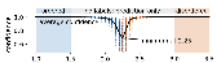
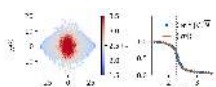
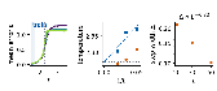
Three Training Regions, Three Different Signals

We train the autoencoder in three temperature windows, $T \leq 1.7$, $T \geq 3.0$ and $2.20 \leq T \leq 2.34$ (shaded in the figure). For each window the top row of the figure shows the mean reconstruction error $\bar{\varepsilon}(T)$, and the bottom row shows T^*, the temperature of the steepest change, against $1/L$.

- **ordered:** $T^* = 2.308, 2.278, 2.264 \to T_c = 2.269$
- **paramagnet:** trained on noise, no structure is learned; for $L = 50$ the error is flat ($T^* = 2.279, 2.253, 2.197$ for $L = 20, 30, 50$)
- **middle:** the jump sits at the edge of the training window, $T^* = 2.389, 2.363, 2.328$

Train on a well-structured, well-understood phase.

Four Ways to Find T_c

	method	what we gave it	what it found
	finite-size scaling	the order parameter m	$T_c = 2.269$ (exact 2.2692)
	CNN (Lecture 4)	labels far from T_c	lowest confidence at 2.25 ($L = 50$, temperature step 0.05)
	PCA	spins only	the magnetisation
	autoencoder	examples of one phase	$T^* = 2.264$ at $L = 50$, sharpening with L

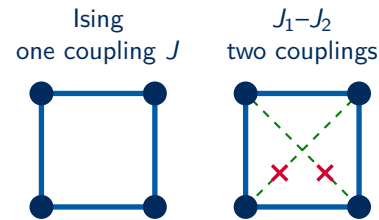
The less we tell the method, the less of the answer comes from our input; explaining the result is still our task.

Model 2

A Frustrated Magnet: the J_1–J_2 Ising Model

Motivation for a Second Model

- for the Ising model we *knew* the answer: one ordered phase, one order parameter m, T_c from Onsager
- in real materials the spins feel **competing interactions**; which order appears, and where, is not known in advance
- **classical model:** $s_i = \pm 1$, each configuration a snapshot of the Boltzmann distribution; quantum models come later in the course

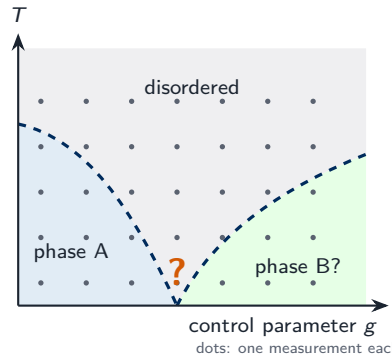


Four parallel spins on a square. The edges ($J_1 > 0$, solid) prefer parallel spins, the diagonals ($J_2 > 0$, dashed) antiparallel ones. No configuration of the four spins satisfies all six bonds.

Mapping an Unknown Phase Diagram Raises Two Problems

a phase diagram is mapped by measuring on a grid of control parameters:

- **which quantity to plot?** the wrong observable hides a phase
- **a dense grid is expensive**, in the lab and on the computer

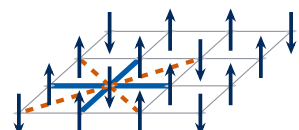


the boundaries (dashed) are unknown; an observable sensitive only to phase A would show phase B as "disordered"; every grid point costs a measurement

The J_1–J_2 Model: Two Couplings that Compete

$$H = -J_1 \sum_{\langle i,j \rangle} s_i s_j \; + \; J_2 \sum_{\langle\langle i,j \rangle\rangle} s_i s_j, \qquad g = \frac{J_2}{J_1}$$

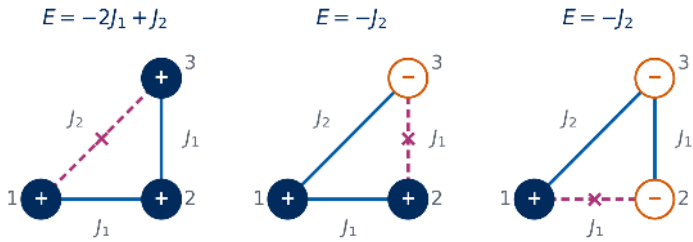
- $J_1 > 0$, 4 nearest neighbours: **ferromagnetic**
- $J_2 > 0$, 4 diagonal neighbours: **antiferromagnetic**
- control parameters g and T: a two-dimensional phase diagram



J_1: nearest J_2: diagonal

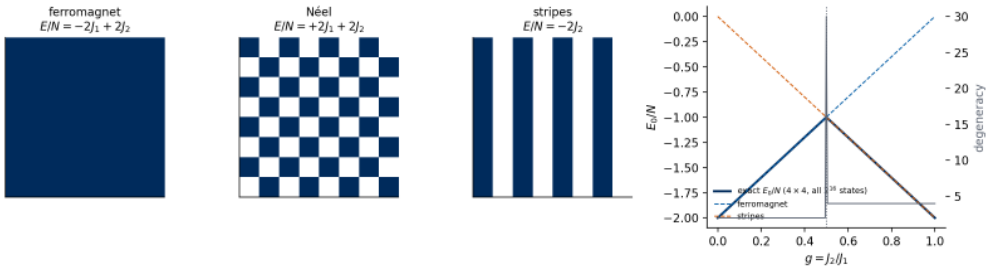
exact limits: $g = 0$ is the Ising model, $T_c = 2.269\, J_1$ (Onsager); for $g \to \infty$ the diagonal bonds split the lattice into two independent Ising antiferromagnets, $T_c \to 2.269\, J_2$

No Configuration Satisfies Every Bond



- In $H_\triangle = -J_1(s_1 s_2 + s_2 s_3) + J_2\, s_1 s_3$ the J_1 bonds favour $s_1 = s_2 = s_3$ and the J_2 bond favours $s_1 = -s_3$, so every state breaks a bond. This is **frustration**.
- Every loop whose coupling signs multiply to -1, here $(+, +, -)$, is frustrated.
- The lattice settles in the **lowest-energy compromise**, ferromagnet for $g < 1/2$, stripes for $g > 1/2$.

Ground States and Frustration



$T = 0$ (bonds per spin: 2 nearest, 2 diagonal): ferromagnet $E/N = -2J_1 + 2J_2$, stripes $E/N = -2J_2$, Néel $E/N = 2J_1 + 2J_2 \Rightarrow$ ferromagnet for $g < 1/2$, stripes for $g > 1/2$. At $g = 1/2$ both couplings cannot be satisfied: **frustration**; the 4×4 torus has 30 ground states instead of $2 + 4$: rows or columns flip at no cost.

Mean Field Misses the Effect of Frustration

Mean field (each spin feels the average of its neighbours):

$$T_{\text{ferro}}^{\text{MF}} = 4(J_1 - J_2), \qquad T_{\text{stripes}}^{\text{MF}} = 4J_2,$$

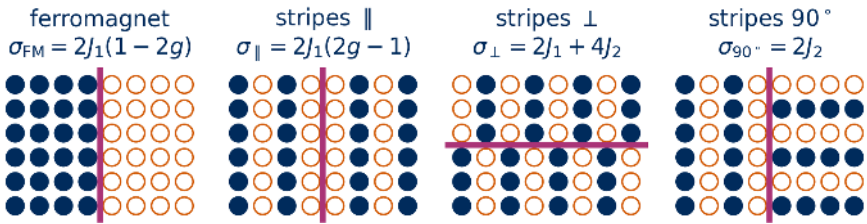
meeting at $g = 1/2$, $T = 2$.

Wrong: fluctuations between the degenerate states destroy the order, and the true $T_c \to 0$ at $g = 1/2$; mean field also gives 4 instead of 2.269 at $g = 0$.

g	0	0.3	0.45	0.7	1.0
mean field T^{MF}	4.000	2.800	2.200	2.800	4.000
transfer matrices of strips, T^*	2.271	1.260	0.711	1.290	2.080

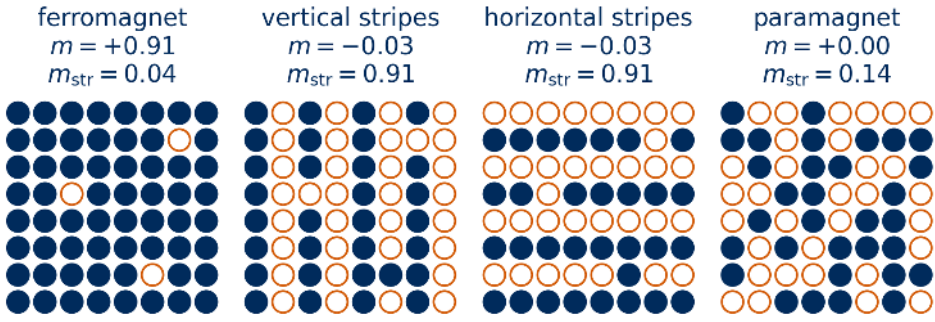
T^*: crossing of ξ_L/L for strip widths 10 and 12 (8 and 10 at $g = 0.45$), Lab 029

Domain Walls Become Free at $g = 1/2$



- σ is the energy per unit length of a straight wall at $T = 0$, obtained by counting bonds (Lab 029). The cheapest walls cost $2J_1|1-2g|$.
- Long walls are suppressed while $\sigma_{\min} > T \ln 3$, a rough Peierls estimate, so the ordering temperature **drops to zero at** $g = 1/2$.
- At $g = 1/2$ whole rows or columns flip at no cost, which gives $2^L + 2^M - 2$ ground states on an $L \times M$ torus.

Three Phases and Their Order Parameters



Fourier component of the spins: $m_{\mathbf{k}} = \frac{1}{N} \sum_{\mathbf{r}} s_{\mathbf{r}}\, e^{i\mathbf{k}\cdot\mathbf{r}}$; $\quad m = m_{(0,0)}$,

$m_{\mathrm{str}} = \left(m^2_{(\pi,0)} + m^2_{(0,\pi)}\right)^{1/2}$

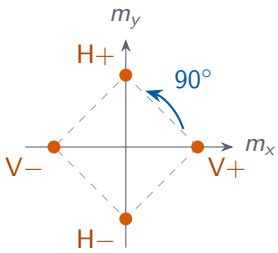
Fourier Components Are the Order Parameters

Fourier component of the spins: $m_{\mathbf{k}} = \frac{1}{N} \sum_{\mathbf{r}} s_{\mathbf{r}}\, e^{i\mathbf{k}\cdot\mathbf{r}}$

phase	$\mathbf{k}$	order parameter	where
ferromagnet	$(0,0)$	$m = \frac{1}{N}\sum_{\mathbf{r}} s_{\mathbf{r}}$	$g < 1/2$, low T
stripes	$(\pi,0),(0,\pi)$	$m_{\mathrm{str}} = \sqrt{m^2_{(\pi,0)} + m^2_{(0,\pi)}}$	$g > 1/2$, low T
paramagnet	–	$m \approx 0,\ m_{\mathrm{str}} \approx 0$	high T

in the stripe phase $m = 0$ **although the system is perfectly ordered**: a scan of m sees it as a paramagnet

Stripe Order Breaks Spin Flip and Lattice Rotation



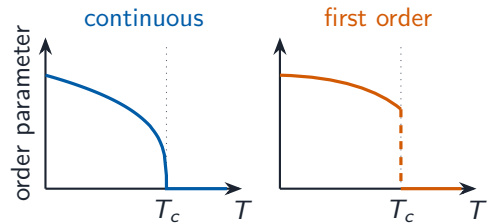
$m_x = m_{(\pi,0)}$, $m_y = m_{(0,\pi)}$

- The ferromagnet has two ordered states, $m = \pm 1$, and breaks only the spin flip.
- The stripes have **four states**, V± and H± (vertical, horizontal), at the corners of a square. Flips, translations and 90° rotations permute them.
- For $g \to \infty$ the sublattices with $x + y$ even and odd decouple, and each orders with its own sign $\sigma_A, \sigma_B = \pm 1$.
- Two coupled Ising variables form the **Ashkin–Teller model**, whose exponents vary along its transition line.

J. Ashkin, E. Teller, Phys. Rev. **64**, 178 (1943), doi:10.1103/PhysRev.64.178; A. Kalz, A. Honecker, M. Moliner, Phys. Rev. B **84**, 174407 (2011), doi:10.1103/PhysRevB.84.174407

Continuous and First-Order Transitions along the Boundary

- **continuous** (second order): the order parameter vanishes without a jump, the correlation length ξ diverges with power laws
- **first order**: the order parameter jumps, two phases coexist at the transition
- $g < 1/2$: continuous transition to the ferromagnet, 2D Ising universality class; $T_c(g) \to 0$ as $g \to 1/2$



Schematic order parameter against temperature. It falls continuously to zero at T_c (left) or jumps to zero (right).

D. P. Landau, K. Binder, Phys. Rev. B **31**, 5946 (1985), doi:10.1103/PhysRevB.31.5946

The Stripe Transition Changes Character along the Line

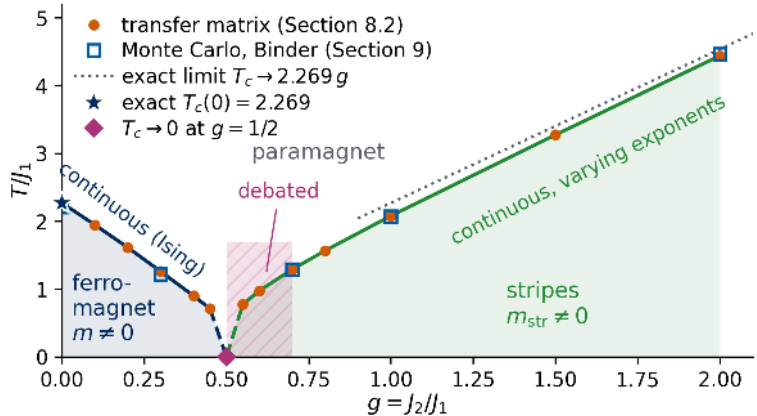
- $g > 1/2$: stripes; weakly first order just above $g = 1/2$, continuous with exponents that change along the line (Ashkin–Teller type) above a point g^*, placed at $g^* \approx 0.67$ by Monte Carlo, lower by tensor networks
- near g^* continuous transitions look first order on lattices of practical size



Decades of work, and the right order parameters, were needed.

A. Kalz, A. Honecker, S. Fuchs, T. Pruschke, Eur. Phys. J. B **65**, 533 (2008), arXiv:0805.0983; A. Kalz, A. Honecker, M. Moliner, Phys. Rev. B **84**, 174407 (2011), doi:10.1103/PhysRevB.84.174407; S. Jin, A. Sen, A. W. Sandvik, Phys. Rev. Lett. **108**, 045702 (2012), arXiv:1110.5874; S. Jin, A. Sen, W. Guo, A. W. Sandvik, Phys. Rev. B **87**, 144406 (2013), arXiv:1212.5339; H. Li, L.-P. Yang, Phys. Rev. E **104**, 024118 (2021), arXiv:2103.09464

Phase Diagram in the (g, T) Plane



exact limits $T_c = 2.269$ $(g = 0)$, $T_c \to 2.269\,g$ $(g \to \infty)$; $T_c \to 0$ at $g = 1/2$, strongest frustration; boundaries: Lab 029 (transfer matrices, Binder cumulants).

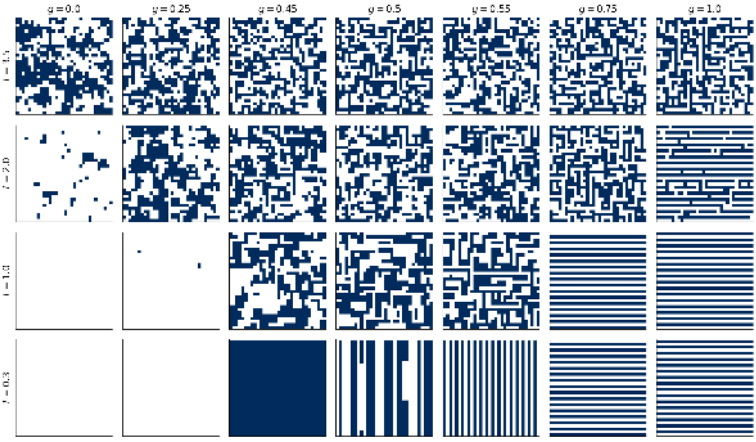
Observables of the J_1–J_2 Model

$\langle O \rangle = \frac{1}{Z} \sum_{\mathbf{s}} O(\mathbf{s})\, e^{-\beta H(\mathbf{s})}$, $Z = \sum_{\mathbf{s}} e^{-\beta H(\mathbf{s})}$, $\beta = 1/T$, with the J_1–J_2 Hamiltonian; n Monte Carlo samples per (g, T), $\overline{x} = \frac{1}{n} \sum_k x_k$; $e = H/N$ with both couplings

	definition through Z	estimator	measures
energy	$\langle e \rangle = -\frac{1}{N} \partial_\beta \ln Z$	$\overline{e}$	bonds
magnetisation	$\langle \lvert m \rvert \rangle$	$\overline{\lvert m \rvert}$	ferromagnetic order
stripe order	$\langle m_{\mathrm{str}} \rangle$	$\overline{m_{\mathrm{str}}}$	stripe order
specific heat	$c = N\beta^2 \left(\langle e^2 \rangle - \langle e \rangle^2 \right)$	$N\beta^2 \left(\overline{e^2} - \overline{e}^{\,2} \right)$	energy fluctuations

$m_{(\pi,0)} = \frac{1}{N} \sum_{\mathbf{r}} (-1)^x s_{\mathbf{r}}$, $m_{(0,\pi)}$ with $(-1)^y$; the larger of the two gives the stripe orientation

Snapshots across the (g, T) Plane

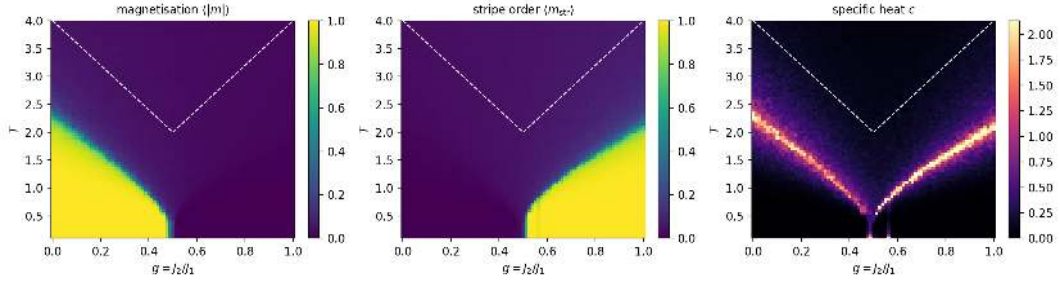


Monte Carlo snapshots $s_{\mathbf{r}} = \pm 1$ of the 32×32 lattice at selected couplings g and temperatures T; our data: a (g, T) grid across all three phases

Metropolis Sampling of the (g, T) Grid

- Metropolis Monte Carlo in JAX: all couplings and chains in one compiled program, each chain cooled from $T = 4$ to $T = 0.1$
- 32×32 lattice; 81 values of g, 79 temperatures, 100 snapshots per point; the whole grid, 639 900 snapshots, took 3567 s of computing time in Lab 018
- high T: noise; low T: uniform ($g < 1/2$) or stripes ($g > 1/2$)
- near $g = 1/2$: ferromagnetic and stripe patches coexist; at $g = 1/2$, $T = 0.3$ the stripes contain doubled rows, one of the many degenerate ground-state patterns

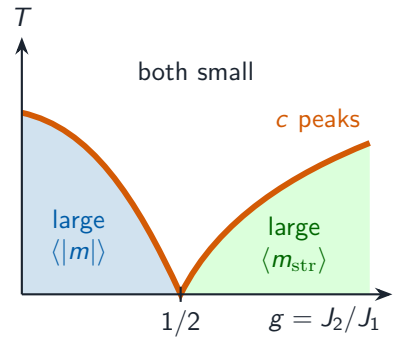
The Observable Decides What We See



$\langle O\rangle = \frac{1}{Z}\sum_{\mathbf{s}} O(\mathbf{s})\, e^{-\beta H(\mathbf{s})}$, $\quad m = \frac{1}{N}\sum_i s_i$, $\quad c = (\langle H^2\rangle - \langle H\rangle^2)/(NT^2)$; dashed: mean field

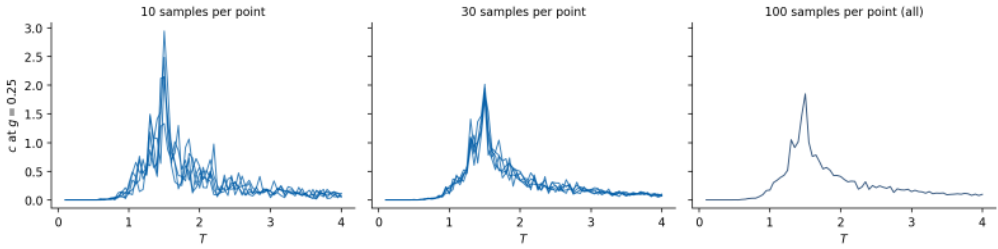
Each Observable Shows a Different Part of the Diagram

- $\langle |m| \rangle$: one ordered region, $g < 1/2$; the stripe phase looks like a paramagnet
- $\langle m_{\mathrm{str}} \rangle$: the second ordered region, visible only because we knew which pattern to measure
- c: peaks at every transition, whatever the order, but it is a variance and needs many samples per point
- dashed lines in the maps: mean-field boundaries



Schematic of the (g, T) plane with the region each observable makes visible.

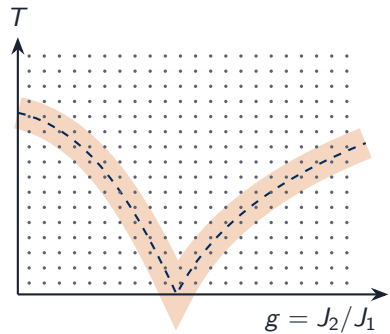
A Dense Scan Is Expensive



- the specific heat at $g = 0.25$ from 10, 30 and all 100 snapshots per temperature; with 10 snapshots the position of the peak jumps between repetitions

A Dense Map Needs 639 900 Configurations

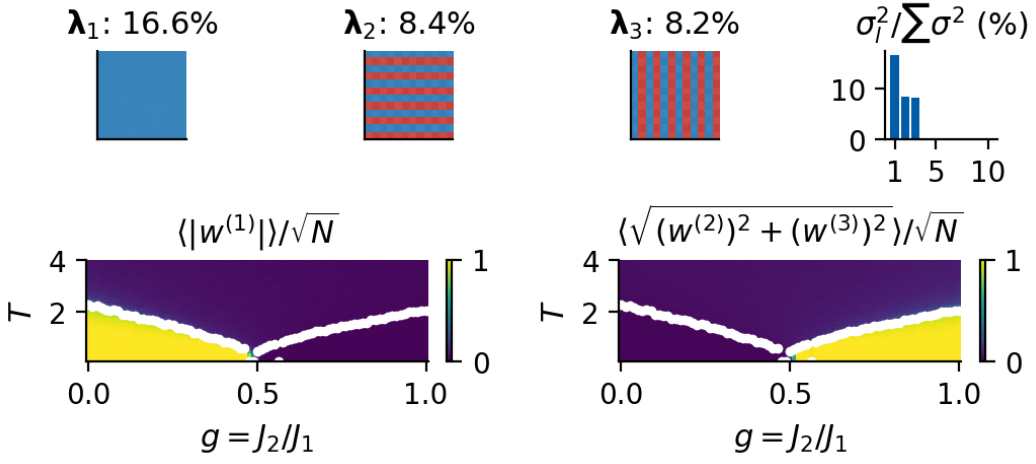
- our map: $81 \times 79 = 6399$ grid points $\times$ 100 snapshots $= 639\,900$ configurations; near a transition consecutive Monte Carlo samples are correlated (critical slowing down), so each costs more
- in a laboratory every grid point is a separate preparation of the sample



Schematic grid with fewer dots than our 6399 points. The boundaries (dashed) occupy a thin band (shaded).

Wanted: a method that needs no order parameter, works from snapshots, and tells us where a dense scan is worth paying for.

PCA Finds Both Order Parameters

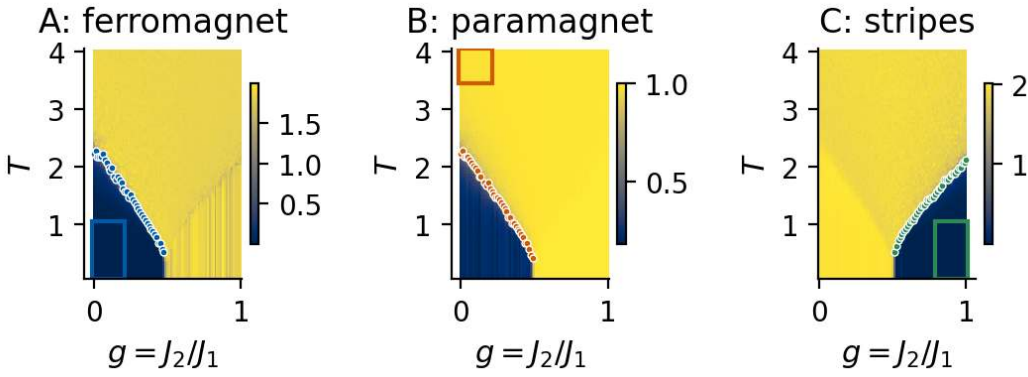


The Principal Components Are the Order Parameters

The top row of the PCA figure shows the principal directions $\boldsymbol{\lambda}_1, \boldsymbol{\lambda}_2, \boldsymbol{\lambda}_3$ on the lattice and the variance fractions $\sigma_l^2 / \sum \sigma^2$. The bottom row maps $\langle |w^{(1)}| \rangle / \sqrt{N}$ and $\langle \sqrt{(w^{(2)})^2 + (w^{(3)})^2} \rangle / \sqrt{N}$ over the (g, T) plane, with $w^{(l)} = \boldsymbol{\lambda}_l^\top (\mathbf{s} - \bar{\mathbf{s}})$.

- $\boldsymbol{\lambda}_1$: uniform ($|w^{(1)}|/\sqrt{N} \approx |m|$); $\boldsymbol{\lambda}_2, \boldsymbol{\lambda}_3$: the two stripe patterns ($\sqrt{(w^{(2)})^2 + (w^{(3)})^2}/\sqrt{N} \approx m_{\mathrm{str}}$); 16.6%, 8.4%, 8.2% of the variance, all others $\leq 0.3\%$
- the maps reproduce $\langle |m| \rangle$ and $\langle m_{\mathrm{str}} \rangle$ (correlations 1.000, 0.9999): PCA found both order parameters, including the one the magnetisation missed
- both are linear in the spins, $m_{\mathbf{k}} = \frac{1}{N} \sum_{\mathbf{r}} s_{\mathbf{r}} e^{i\mathbf{k}\cdot\mathbf{r}}$; order that is not linear in the spins is out of reach of PCA

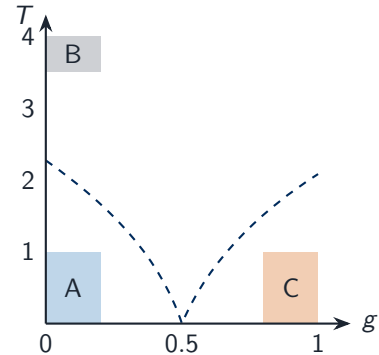
Reconstruction-Error Maps for Three Training Regimes



- convolutional autoencoder (4 latent numbers), trained on one corner only (rectangle), with every snapshot also flipped, $\mathbf{s} \to -\mathbf{s}$: the symmetry of H

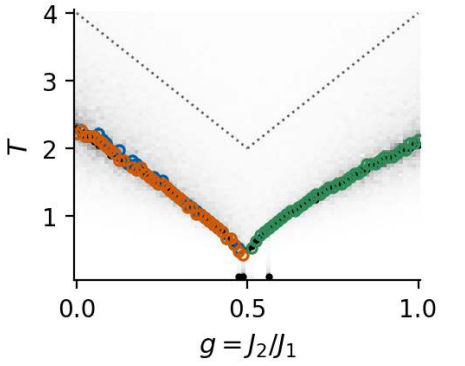
The Three Training Regimes Rebuild Different Phases

- **A, ferromagnet** ($g \leq 0.2,\ T \leq 1$): the ferromagnetic phase is rebuilt almost exactly; stripes and paramagnet are both "not ferromagnet"
- **B, paramagnet** ($g \leq 0.2,\ T \geq 3.5$): it learns little more than the average; the ferromagnet is rebuilt *better* than its own data, the stripe phase is **invisible**
- **C, stripes** ($g \geq 0.8,\ T \leq 1$): the stripe phase and its boundary



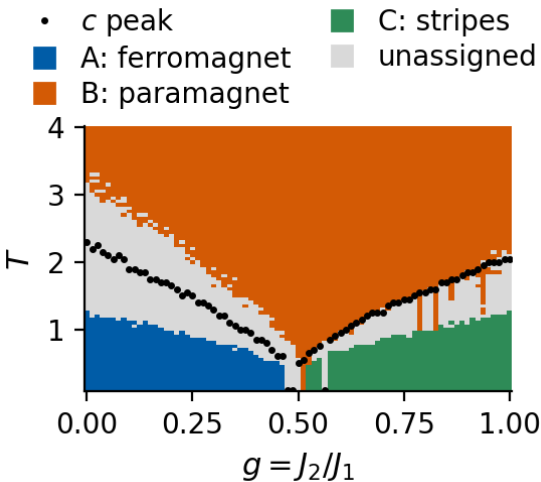
training regions A, B, C in the (g, T) plane; dashed: sketch of the transition line

Each Regime Sees the Boundary of Its Own Phase



- boundary point: for every g, the temperature where the mean error changes fastest
- A and B: the ferromagnetic line, $0 \leq g \leq 0.48$ (A), 0.49 (B); C: the stripe line, $0.51 \leq g \leq 1$
- median distance from the specific-heat peak: 0.025, half the temperature step of the grid
- one training region answers one question, *is this like my data?*; it cannot tell two different anomalies apart

Combining the Regimes into a Phase Map



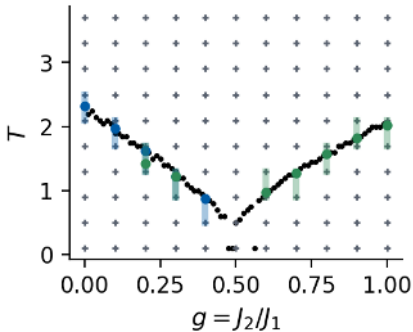
- anomaly score of regime k: $a_k = |\bar{\varepsilon}_k - \mu_k|/\sigma_k$, the distance of the error from its level on the training region; a point goes to the regime with the smallest score, *unassigned* if all are above 3
- the three colours reproduce the three phases

Unassigned Points Lie on the Ordered Side of the Transitions

- **unassigned** (21% of the grid): mostly the ordered side of each transition, where fluctuations make snapshots atypical for a model trained deep in the phase
- around $g = 1/2$ at low T: *paramagnet*, consistent with $T_c \to 0$ at $g = 1/2$: no order within our temperature range

assigned to	A, ferromagnet	B, paramagnet	C, stripes	unassigned
fraction of the 81×79 grid	9.9%	59.2%	10.1%	20.7%

A Coarse Scan Tells Where to Measure



- coarse grid: $\Delta g = 0.1$, $\Delta T = 0.4$, 10 snapshots per point; regimes A and C trained on the coarse data only
- bars: temperature brackets where the error jumps; dots: boundary after measuring densely *inside the brackets only*
- the dots land on the specific-heat ridge of the dense scan
- cost: $1100 + 2160 = 3260$ snapshots, **0.5%** of the 639 900 used for the dense specific-heat map

Results and Requirements of the Methods

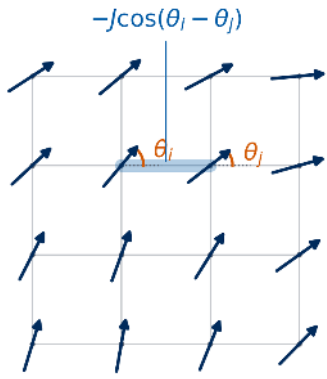
method	input	result vs. the known physics
$\langle \lvert m \rvert \rangle$ scan	the ferromagnetic order parameter	misses the stripe phase
specific heat	many samples per point	both boundaries; expensive
PCA	all snapshots	uniform and stripe patterns: both order parameters
AE, one regime	snapshots of one corner	the boundary of that phase only
AE, three regimes	three corners	three phases; a coarse scan at 0.5% of the cost

The method finds where something changes; which order sets in there is still the physicist's question.

Model 3

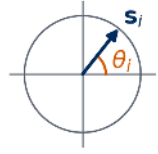
The 2D XY Model

Planar Spins and the XY Hamiltonian



$$H = -J \sum_{\langle i,j \rangle} \cos(\theta_i - \theta_j)$$

planar spins $\mathbf{s}_i = (\cos\theta_i, \sin\theta_i)$
$\theta_i \in [0, 2\pi)$, $J > 0$, $\langle i,j \rangle$: neighbours



parallel: $-J$
antiparallel: $+J$
small angle δ: $-J + \frac{J}{2}\delta^2$

$L \times L$ square lattice, periodic boundaries, $N = L^2$ spins; $\langle i,j \rangle$: the $2N$ nearest-neighbour bonds, each counted once; we set $J = 1$, $k_B = 1$

Configurations and Magnetisation

- spin i: a unit vector in the plane, $\mathbf{s}_i = (\cos\theta_i, \sin\theta_i)$, angle $\theta_i \in [0, 2\pi)$; a configuration is the set of all angles $\boldsymbol{\theta} = (\theta_1, \ldots, \theta_N)$, with energy $H(\boldsymbol{\theta}) = -J \sum_{\langle i,j \rangle} \cos(\theta_i - \theta_j)$
- **magnetisation** of one configuration, a two-component vector $\mathbf{m} = (m_x, m_y)$:

$$m_x(\boldsymbol{\theta}) = \frac{1}{N} \sum_{i=1}^{N} \cos\theta_i, \qquad m_y(\boldsymbol{\theta}) = \frac{1}{N} \sum_{i=1}^{N} \sin\theta_i$$

- its length depends only on angle differences; its direction turns by α when all angles turn by α:

$$|\mathbf{m}|^2 = m_x^2 + m_y^2 = \frac{1}{N^2} \sum_{i,j} \cos(\theta_i - \theta_j), \qquad 0 \le |\mathbf{m}| \le 1$$

- all angles equal: $|\mathbf{m}| = 1$; independent uniform random angles: $\langle |\mathbf{m}|^2 \rangle = 1/N$

Thermal Averages and the Mean Magnetisation

- **thermal average** at temperature T: the angles are continuous, so the sum over configurations is an integral,

$$\langle O \rangle = \frac{1}{Z} \int_0^{2\pi} d\theta_1 \cdots \int_0^{2\pi} d\theta_N \; O(\boldsymbol{\theta})\, e^{-\beta H(\boldsymbol{\theta})}, \qquad Z = \int d\boldsymbol{\theta}\; e^{-\beta H(\boldsymbol{\theta})}, \quad \beta = 1/T$$

- Monte Carlo draws configurations $\boldsymbol{\theta}^{(k)}$ with probability density $e^{-\beta H}/Z$ and replaces the integral by the sample mean, $\langle O \rangle \approx \frac{1}{n} \sum_{k=1}^{n} O(\boldsymbol{\theta}^{(k)})$

- a global rotation $\theta_i \to \theta_i + \alpha$ leaves H unchanged and rotates $\mathbf{m}$ by α, so $\langle \mathbf{m} \rangle = 0$ on every finite lattice

- we therefore measure $\langle |\mathbf{m}| \rangle$ (or $\langle |\mathbf{m}|^2 \rangle$): nonzero on a finite lattice, $\to 0$ as $L \to \infty$ at every $T > 0$ (Mermin–Wagner)

- example, $T = 0.3$ (Lab 030): $\langle |\mathbf{m}| \rangle = 0.940,\ 0.907,\ 0.874$ for $L = 8,\ 32,\ 128$

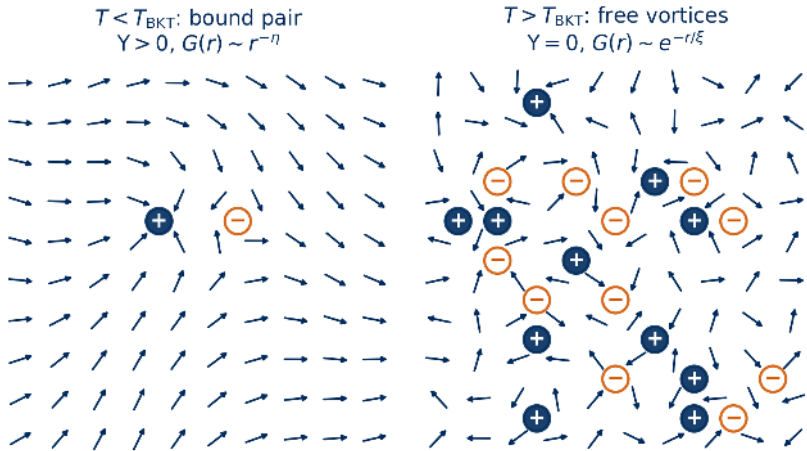
A Continuous Symmetry, Realised in Films and Magnets

The phase of a superfluid or superconducting film and the direction of a planar magnet are angles, exactly like θ_i.

- **symmetry:** H is unchanged by a global rotation $\theta_i \to \theta_i + \alpha$ for every angle α, a continuous symmetry (with the reflection $\theta_i \to -\theta_i$: the group $O(2)$)
- **ground states:** all spins parallel, $E_0 = -2JN$; a continuous family labelled by one angle (Ising: two ground states)
- **realisations:** ^{4}He films (superfluid density measured by Bishop and Reppy, 1978), thin superconducting films, planar magnets
- **Nobel Prize in Physics 2016** (Thouless; Haldane, Kosterlitz): "for theoretical discoveries of topological phase transitions and topological phases of matter"

D. J. Bishop, J. D. Reppy, Phys. Rev. Lett. **40**, 1727 (1978), doi:10.1103/PhysRevLett.40.1727; nobelprize.org/prizes/physics/2016/summary; derivations: Lab 030

Two Phases without a Local Order Parameter



low T: neighbours nearly aligned, the direction drifts slowly, vortices only in bound pairs; high T: directions random beyond a few spacings, free vortices ($+$, $-$: winding ± 1)

Stiffness and Correlations Replace the Order Parameter

	$T < T_{\mathrm{BKT}}$: quasi-long-range order	$T > T_{\mathrm{BKT}}$: disorder
correlations $G(r)$	power law $r^{-\eta(T)}$	exponential $e^{-r/\xi}$
helicity modulus Υ	$\Upsilon > 0$	$\Upsilon = 0$
vortices	bound in pairs	free
magnetisation, $L \to \infty$	0	0

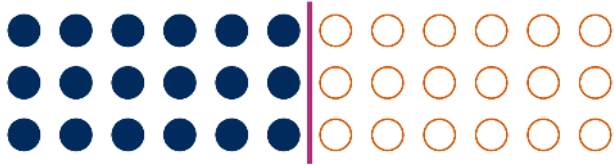
$G(r) = \langle \cos(\theta_0 - \theta_r) \rangle$: correlation at distance r; ξ: correlation length; Υ: free-energy cost of a twist Δ, $F(\Delta) - F(0) = \frac{\Upsilon}{2}\Delta^2$

- no broken symmetry (Mermin–Wagner); $T_{\mathrm{BKT}} = 0.8929\,J$
- Berezinskii–Kosterlitz–Thouless (BKT) transition, **infinite order**: $\xi \propto e^{b/\sqrt{t}}$, $t = T - T_{\mathrm{BKT}}$, b: a non-universal constant; jump $\Upsilon(T_{\mathrm{BKT}}^{-}) = 2\,T_{\mathrm{BKT}}/\pi$
- 2D Ising, second order: $\xi \propto |t|^{-\nu}$, $\nu = 1$

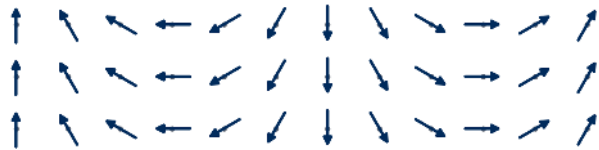
M. Hasenbusch, J. Phys. A **38**, 5869 (2005), doi:10.1088/0305-4470/38/26/003

A Slow Twist Costs Almost No Energy

(a) Ising: sharp domain wall, $2J$ per broken bond



(b) XY: spin wave, 2π over ℓ sites, $\frac{J}{2}(2\pi/\ell)^2$ per bond



turning the spins by 2π over ℓ sites costs $\frac{J}{2}(2\pi/\ell)^2$ per bond, about ℓ^{d-2} in total: independent of ℓ in $d = 2$; an Ising domain wall costs $\propto \ell^{d-1}$

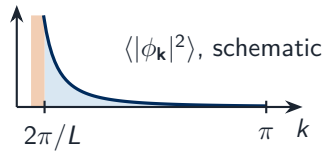
Spin Waves Destroy the Magnetisation

Expanding $\cos x \approx 1 - x^2/2$ gives independent Gaussian modes (spin waves):

$$\langle \phi^2 \rangle \simeq \frac{T}{2\pi J} \ln L, \qquad \langle |\mathbf{m}| \rangle \propto L^{-T/(4\pi J)} \to 0, \qquad G(r) \propto r^{-\eta}, \quad \eta = \frac{T}{2\pi J}$$

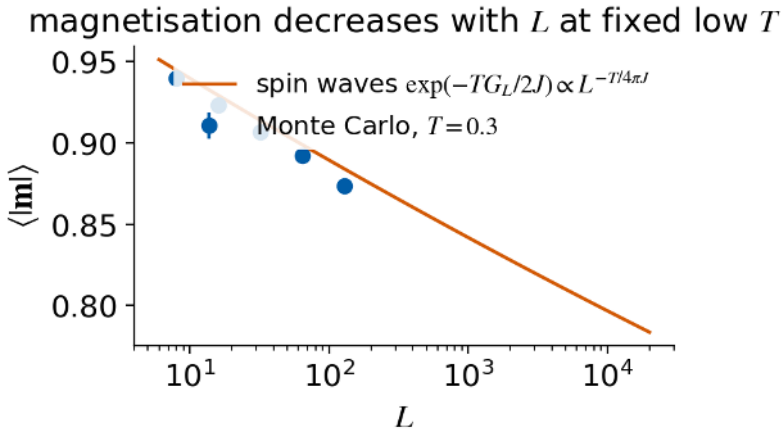
$\phi_i = \theta_i - \bar{\theta}$: angle from the mean direction; $\mathbf{m} = \frac{1}{N} \sum_i (\cos\theta_i, \sin\theta_i)$

- mode $\mathbf{k}$ costs $\frac{J}{2} k^2 |\phi_{\mathbf{k}}|^2$, so equipartition gives $\langle |\phi_{\mathbf{k}}|^2 \rangle = T/(Jk^2)$
- $\langle \phi^2 \rangle = \frac{1}{N} \sum_{\mathbf{k}} \frac{T}{Jk^2} \simeq \frac{T}{2\pi J} \int_{2\pi/L}^{\pi} \frac{dk}{k}$ grows as $\ln L$, carried by the long waves near $k = 2\pi/L$
  - no long-range order at $T > 0$ (Mermin–Wagner)
  - power-law $G(r)$: **quasi-long-range order**
  - **finite-size magnetisation** on every lattice



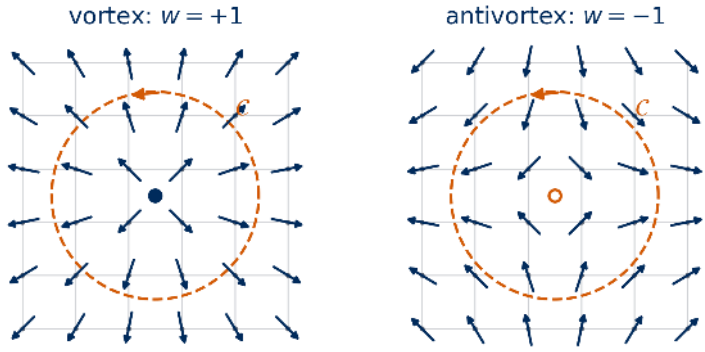
N. D. Mermin, H. Wagner, Phys. Rev. Lett. **17**, 1133 (1966), doi:10.1103/PhysRevLett.17.1133

The Magnetisation Decays Slowly with the Lattice Size



$T = 0.3$: fitted exponent 0.0261 (theory 0.0239); halving $\langle |\mathbf{m}| \rangle$ needs $L \times 3.5 \cdot 10^{11}$

Vortices Are Topological Defects



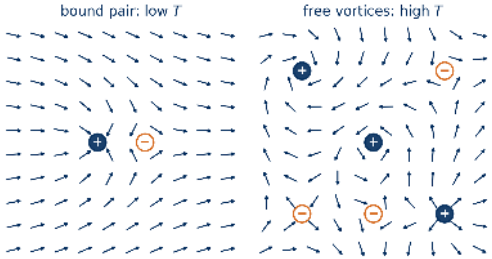
$w = \frac{1}{2\pi}\oint_C \nabla\theta \cdot d\mathbf{l}$ = number of turns of the spins along C

lattice: $w = \frac{1}{2\pi}\sum_{\square}[\theta_j - \theta_i] \in \mathbb{Z}$, the four bonds of a plaquette counter-clockwise, $[x]$: angle difference mapped to $[-\pi, \pi)$, so $w \in \{-1, 0, 1\}$ (-2 has probability zero); on a torus $\sum w = 0$

Entropy Unbinds the Vortices

Because $|\nabla\theta| = 1/r$ around a vortex, its energy grows with the size; its centre can sit on any of $(L/a)^2$ plaquettes:

$$E_v = \pi J \ln \frac{L}{a}, \qquad S_v = 2 \ln \frac{L}{a}, \qquad F_v = E_v - TS_v = (\pi J - 2T) \ln \frac{L}{a}$$



a: lattice spacing

- $T < \pi J/2$: only bound pairs; above: free vortices
- $T_{\mathrm{BKT}} = 0.893$ lies below $\pi J/2 = 1.571$ because bound pairs **screen** the interaction
- vortex density at $T = 0.90$: 0.009, smooth in T

J. M. Kosterlitz, D. J. Thouless, J. Phys. C **6**, 1181 (1973), doi:10.1088/0022-3719/6/7/010

Bound Vortex Pairs Are Present at Every Temperature

Far from a vortex–antivortex pair at distance d the two windings cancel, so the energy of the pair does not grow with L,

$$E_{\mathrm{pair}}(d) \approx 2\pi J \ln \frac{d}{a} + 2E_{\mathrm{core}}.$$

Here a is the lattice spacing and E_{core} the energy of one vortex core.

- On the lattice (Lab 030) the slope $dE_{\mathrm{pair}}/d\ln d$ is 6.40 for $d = 8 \to 16$ and 6.31 for $d = 16 \to 32$, close to $2\pi J = 6.28$.
- A pair has a finite energy, so pairs are thermally activated. The vortex density follows $n_v \propto e^{-E_a/T}$ with a fitted $E_a \approx 7.3\,J$ ($T \le 0.75$, $L = 48$).

The low-temperature phase contains bound pairs at every $T > 0$.

Screening by Bound Pairs Sets the Universal Jump

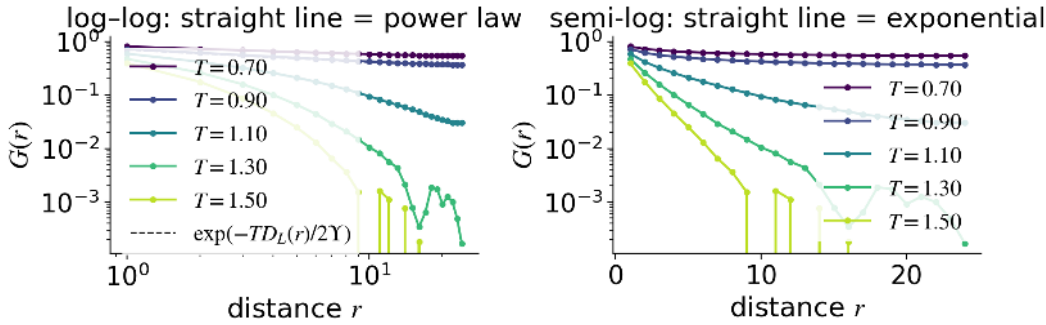
By analogy with electrostatics, vortices act like 2D charges with a logarithmic interaction, and bound pairs act like dipoles in a dielectric that **screen** it. The coupling J in the vortex free energy becomes the stiffness $\Upsilon < J$,

$$F_v = (\pi\Upsilon - 2T)\ln\frac{L}{a}.$$

- Free vortices appear once Υ falls to $2T/\pi$. This condition gives the **universal jump** $\Upsilon(T_{\mathrm{BKT}}^{-}) = 2T_{\mathrm{BKT}}/\pi$ and lowers the transition from $\pi J/2 = 1.571$ to $T_{\mathrm{BKT}} = 0.893$.
- Monte Carlo at $T = 0.70$, $L = 48$ gives $\Upsilon = 0.777$, below the spin-wave value $J - T/4 = 0.825$.

J. M. Kosterlitz, J. Phys. C **7**, 1046 (1974), doi:10.1088/0022-3719/7/6/005; D. R. Nelson, J. M. Kosterlitz, Phys. Rev. Lett. **39**, 1201 (1977), doi:10.1103/PhysRevLett.39.1201

Power-Law Correlations below, Exponential above

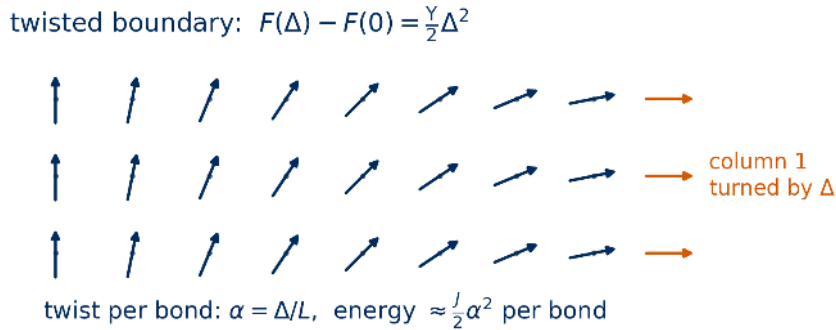


$G(r) = \frac{1}{N}\sum_i \cos(\theta_i - \theta_{i+r})$, $L = 48$

- below: $r^{-\eta}$, $\eta = T/(2\pi\Upsilon)$, spin waves within 1% for $T \leq 0.8$; $\eta(T_{\mathrm{BKT}}) = 1/4$
- above: $e^{-r/\xi}$, $\xi = 9.3$ at $T = 1.05$, 1.5 at $T = 1.50$

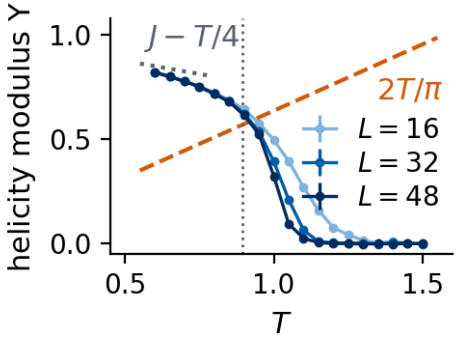
J. M. Kosterlitz, J. Phys. C **7**, 1046 (1974), doi:10.1088/0022-3719/7/6/005

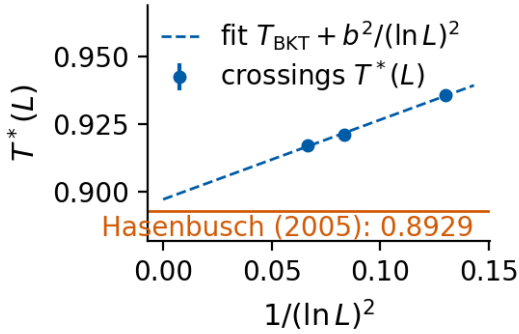
The Stiffness Is the Response to a Twist



A phase with order resists a twist, as a superfluid resists a phase gradient: $\Upsilon = \frac{1}{N}\partial_\alpha^2 F|_{\alpha=0}$, $\alpha = \Delta/L$ the twist per bond, $F = -T \ln Z$; measured in equilibrium from bond fluctuations

The Stiffness Jump Locates T_{BKT}



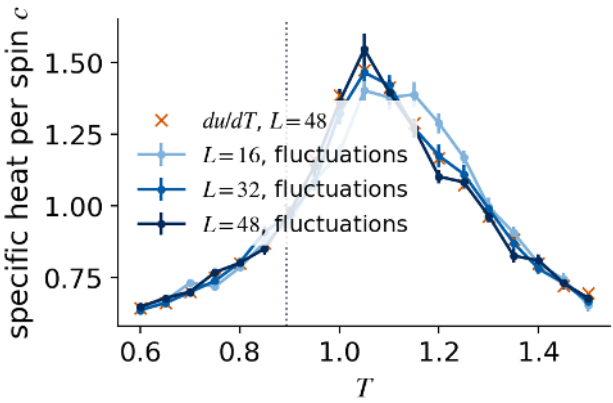


- universal jump $\Upsilon(T_{\mathrm{BKT}}^{-}) = 2T_{\mathrm{BKT}}/\pi$ (Nelson–Kosterlitz); left: $\Upsilon_L(T)$ and $2T/\pi$
- crossings $T^*(L) = 0.936,\ 0.921,\ 0.917$ ($L = 16, 32, 48$); $\xi(T^*) \approx L$ gives $T^*(L) = T_{\mathrm{BKT}} + b^2/(\ln L)^2$; fit (right): 0.897 ± 0.003, literature 0.8929

D. R. Nelson, J. M. Kosterlitz, Phys. Rev. Lett. **39**, 1201 (1977), doi:10.1103/PhysRevLett.39.1201

The Specific Heat Peaks above T_{BKT}

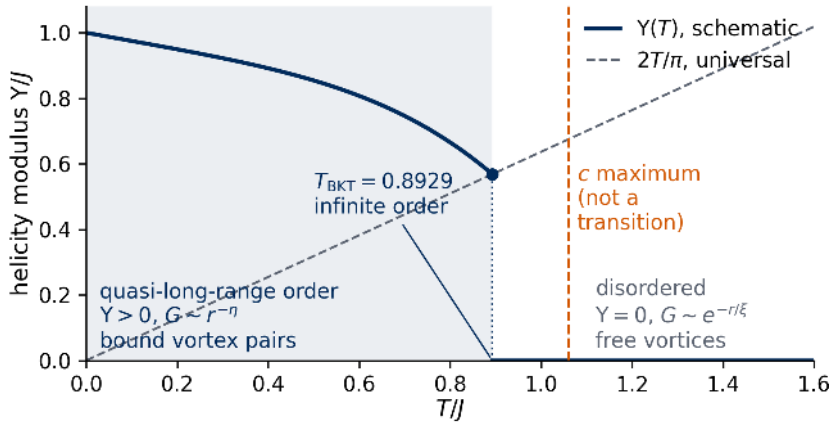
$c = (\langle H^2 \rangle - \langle H \rangle^2)/(NT^2)$: specific heat per spin



dotted: T_{BKT}

- singular free energy $\propto \xi^{-2} \propto e^{-2b/\sqrt{t}}$: all derivatives vanish at $t \to 0^+$; no divergence
- maximum at $T = 1.12,\ 1.07,\ 1.06$ $(L = 16, 32, 48)$: **not a transition**
- **warning:** a detector based on the energy finds this maximum

The Phase Diagram on the Temperature Axis



schematic $\Upsilon(T)$; T_{BKT}: infinite-order transition; the c maximum lies in the disordered phase

Reference Temperatures for the Comparison

Every temperature an unsupervised method returns is compared with these values.

reference	value	meaning
T_{BKT} (Monte Carlo, Hasenbusch)	0.8929	the transition
stiffness crossings $T^*(L)$	0.936, 0.921, 0.917 $\rightarrow$ 0.897(3)	estimates of T_{BKT}
specific-heat maximum	1.12, 1.07, 1.06	not a transition
estimate $\pi J/2$	1.571	no screening

values for $L = 16, 32, 48$; data for the machine learning: Monte Carlo snapshots of Lab 030, $L = 16, 32, 48$, 19 temperatures 0.60–1.50, 192 snapshots per temperature, no labels

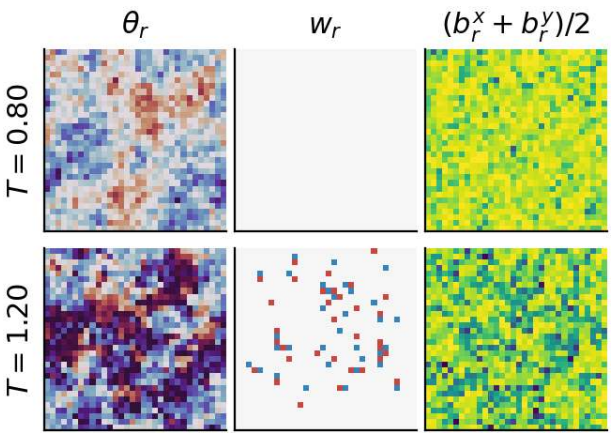
Thermodynamic Observables: Definition and Estimator

$Z = \int d\boldsymbol{\theta}\, e^{-\beta H}$, $\beta = 1/T$, $\langle O \rangle = Z^{-1} \int d\boldsymbol{\theta}\, O\, e^{-\beta H}$; estimator: sample mean $\overline{x} = \frac{1}{n} \sum_k x_k$ over $n = 24$ chains $\times$ 1000 Monte Carlo steps per T

	definition through Z	**estimator**
energy	$u = -\frac{1}{N} \partial_\beta \ln Z$	$\overline{E}/N$
specific heat	$c = \frac{\beta^2}{N} \partial_\beta^2 \ln Z$	$\frac{\beta^2}{N} \left(\overline{E^2} - \overline{E}^2 \right)$
magnetisation	$\langle \lvert \mathbf{m} \rvert \rangle$	$\overline{\lvert \mathbf{m} \rvert}$
helicity modulus	$\Upsilon = \frac{1}{N} \partial_\alpha^2 F \lvert_0$, $F = -T \ln Z(\alpha)$	$\frac{1}{2N} \left[\overline{C_x + C_y} - \beta \, \overline{S_x^2 + S_y^2} \right]$
vortex density	$\langle n_v \rangle$	$\overline{n_v}$

E: energy of a step; $C_\mu = \sum_r \cos \Delta_r^\mu$, $S_\mu = \sum_r \sin \Delta_r^\mu$, $\Delta_r^\mu = \theta_{r+\hat{\mu}} - \theta_r$; α: twist added to every Δ_r^x; n_v: fraction of plaquettes with $w \neq 0$; variances divide by n

Three Representations of a Snapshot



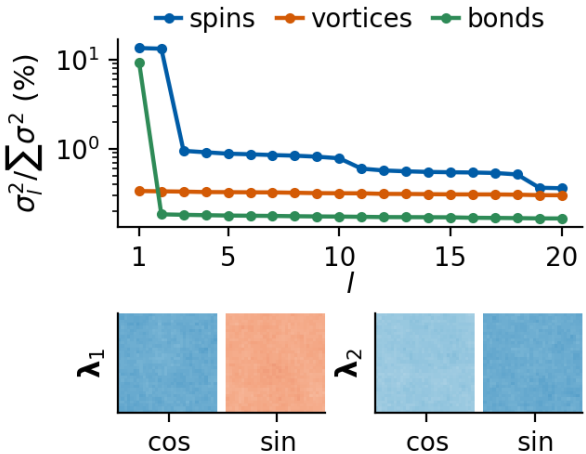
one $L = 32$ snapshot per row

The input decides what an algorithm can see.

- **spins** $(\cos\theta_r, \sin\theta_r)$: everything
- **vortices** w_r: winding number per plaquette, unrelated to the PCA projections $w^{(l)}$
- **bonds** $b_r^\mu = \cos(\theta_{r+\hat{\mu}} - \theta_r)$: local energy
- every snapshot: random global rotation α, the symmetry of H

M. J. S. Beach, A. Golubeva, R. G. Melko, Phys. Rev. B **97**, 045207 (2018), doi:10.1103/PhysRevB.97.045207

PCA Spectra of the Three Representations

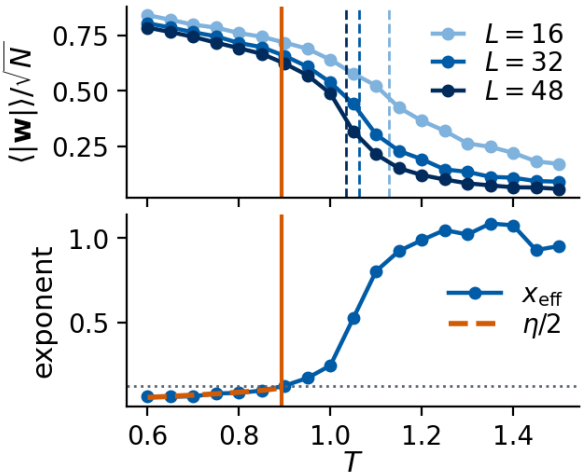


Principal component analysis (PCA) finds the directions $\boldsymbol{\lambda}_l$ of largest variance, with variance fractions $\sigma_l^2/\sum\sigma^2$. We use all snapshots of $L = 32$.

- **spins:** $\boldsymbol{\lambda}_{1,2}$ carry 13.4, 13.2%, uniform: (m_x, m_y)
- **bonds:** 9.3%: energy
- **vortices:** flat, no linear pattern

W. Hu, R. R. P. Singh, R. T. Scalettar, Phys. Rev. E **95**, 062122 (2017), doi:10.1103/PhysRevE.95.062122

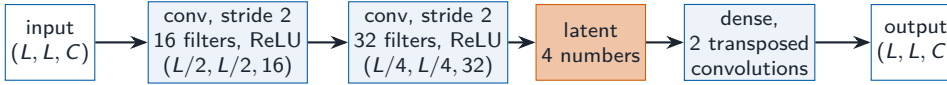
PCA Finds the Finite-Lattice Magnetisation



The top panel shows $\langle|\mathbf{w}|\rangle/\sqrt{N} = \langle|\mathbf{m}|\rangle$, with $\mathbf{w} = (w^{(1)}, w^{(2)})$ and $w^{(l)} = \boldsymbol{\lambda}_l^\top(\mathbf{x} - \bar{\mathbf{x}})$. The bottom panel shows the exponent x_{eff} in $\langle|\mathbf{w}|\rangle \propto L^{-x_{\text{eff}}}$ together with $\eta/2$.

- the finite-size magnetisation (spin-wave frame)
- steepest fall $T_{\text{PCA}} = 1.13,\ 1.06,\ 1.03$ $(L = 16, 32, 48)$: **drifts with L**

Convolutional Autoencoder on the Torus

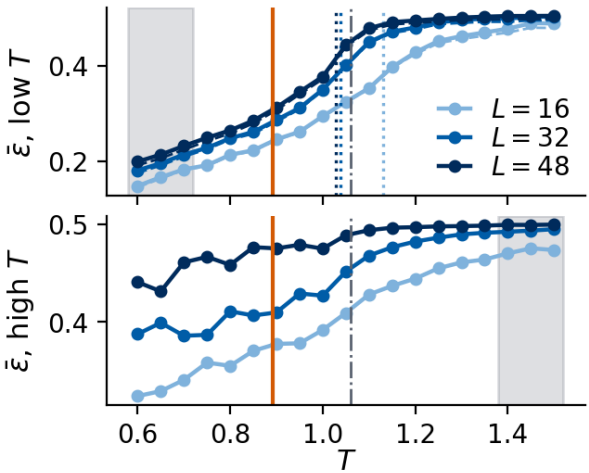


An autoencoder (AE) compresses a snapshot to a few numbers and reconstructs it; trained on one well-understood phase, it measures at every T how much the snapshots differ from that phase.

- periodic padding (the lattice is a torus); C input channels; ReLU $= \max(0, x)$
- **training:** $T \leq 0.70$, 288 snapshots of chains 0–11, random rotations; Adam optimiser, 40 epochs, two seeds
- **reconstruction error** $\varepsilon(\mathbf{x}) = \frac{1}{NC} \sum_{r,c} (x_{r,c} - \hat{x}_{r,c})^2$; $\bar{\varepsilon}(T)$: mean over 96 held-out snapshots (chains 12–23)
- **estimate** $T_{\mathrm{AE}}(L)$: steepest rise of $\bar{\varepsilon}(T)$; error: bootstrap over chains

K. Kottmann, P. Huembeli, M. Lewenstein, A. Acín, Phys. Rev. Lett. **125**, 170603 (2020), doi:10.1103/PhysRevLett.125.170603

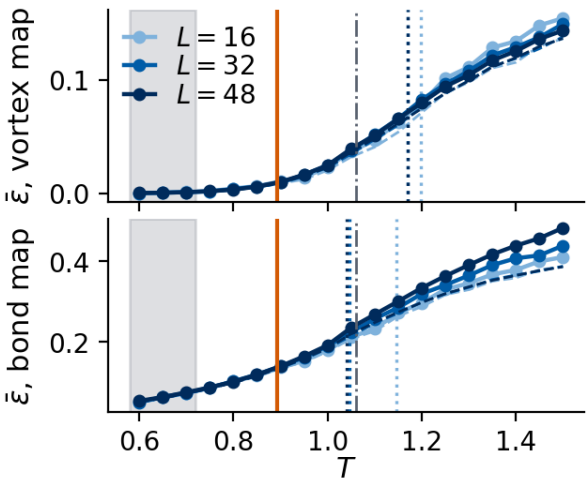
On Raw Spins the Autoencoder Learns the Magnetisation



dashed: $(1 - |\mathbf{m}|^2)/2$, error of $\hat{\mathbf{s}}_r = \mathbf{m}$; orange: T_{BKT}; grey: training T

- **low-T training:** error = magnetisation baseline; $T_{\mathrm{AE}} = 1.13,\ 1.04,\ 1.03$
- **high-T training:** ordered snapshots look *less* anomalous

The Input Representation Decides the Signal



dashed: $\frac{1}{N}\sum_r w_r^2 \approx n_v$ (top), variance of b_r (bottom)

- **vortices:** error $= n_v$; $T_{\mathrm{AE}} = 1.20,\ 1.17,\ 1.17,$ **above the c maximum**
- **bonds:** $T_{\mathrm{AE}} = 1.15,\ 1.05,\ 1.04$: the c maximum
- counts defects, not whether they are bound

Larger Networks Give the Same Temperature

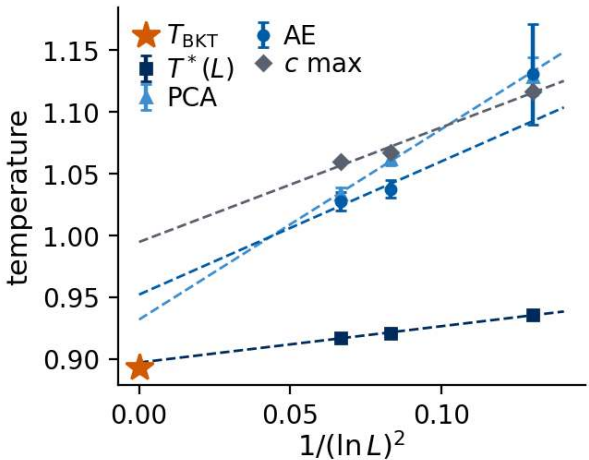
autoencoder on raw spins, $L = 32$, one change at a time, two seeds each; final training loss and T_{AE} bootstrap error

	baseline	latent 2	latent 8	latent 16	width $\times 2$	width $\times 4$	epochs $\times 3$
loss	0.194	0.209	0.187	0.173	0.192	0.189	0.184
T_{AE}	1.038(7)	1.037(7)	1.037(8)	1.039(8)	1.036(7)	1.038(7)	1.038(7)

baseline: 4 latent numbers, 16 and 32 filters, 40 epochs; width $\times k$: all filter numbers multiplied by k

- the loss changes, the temperature does not: largest shift 0.14 bootstrap errors (vortex AE: 0.10)
- **within these variants, capacity and training time do not move T_{AE}**

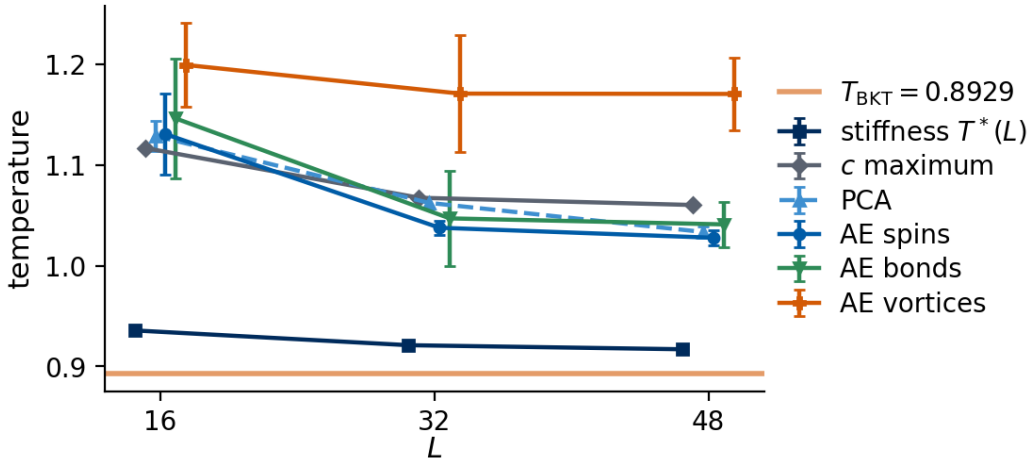
The Magnetisation Crossover Extrapolates towards T_{BKT}



fit $T(L) = T_\infty + b^2/(\ln L)^2$, from $\xi(T) \approx L$

- T^*: 0.897 ± 0.003; PCA: 0.932 ± 0.020; AE: 0.952 ± 0.036
- $T_\infty = 0.8929$ fixed: $p = 0.12,\ 0.13$: T_{BKT} not excluded
- c max: 0.995
- **3 sizes; the form is BKT physics**

Unsupervised Estimates against the Known Temperatures



$L = 48$: unsupervised estimates 1.03–1.17, **0.14–0.28 above** T_{BKT}; only $T^*(L)$ is close

Snapshots Reveal the Crossover, Not T_{BKT}

method	measures	T, $L = 48$	vs. the known physics
stiffness crossing	$\Upsilon_L = 2T/\pi$	0.917	$\to 0.897(3)$, $T_{\mathrm{BKT}} = 0.8929$
specific heat	maximum of c	1.06	smooth, above T_{BKT}
PCA on spins	finite-size $\lvert\mathbf{m}\rvert$	1.03	crossover $\to 0.932 \pm 0.020$
AE on spins	finite-size $\lvert\mathbf{m}\rvert$	1.03	crossover $\to 0.952 \pm 0.036$
AE on bonds	energy fluctuations	1.04	specific-heat maximum
AE on vortices	vortex density	1.17	smooth proliferation

An unsupervised detector finds what varies most in the snapshots; the physics tells us what that is and how to extrapolate it.

Three Models, One Protocol

model	order	what the raw-snapshot detector sees	the input or quantity that works
2D Ising	magnetisation m, linear in the spins	PCA: PC 1 $= m$; autoencoder: $T^* \to T_c$ (2.264 at $L = 50$, exact 2.269)	raw spins
J_1–J_2 Ising	m and stripe order m_{str}, both linear in the spins	PCA: both order parameters; autoencoder: the boundary of its training phase	raw spins, several training regions
2D XY	no local order parameter; quasi-long-range order, vortex unbinding at $T_{\mathrm{BKT}} = 0.893$	PCA and autoencoder on spins: the finite-lattice magnetisation crossover, ≈ 1.03 at $L = 48$; on bond energies: the specific-heat maximum; on vortex maps: ≈ 1.17	only the stiffness Υ locates T_{BKT}

Summary

- **unsupervised learning** finds structure in data without labels: a few coordinates, groups, unusual samples (PCA, t-SNE, k-means, DBSCAN, autoencoders)
- **anomaly detection** with an autoencoder trained on one region locates phase boundaries without labels and without an order parameter
- **2D Ising:** PCA finds the magnetisation; the autoencoder signal converges to T_c with L when trained on the ordered phase; the training region decides the signal

2D Ising	temperature
exact (Onsager)	2.269
χ peaks, $L \to \infty$	2.269
Binder crossings	2.250, 2.278
autoencoder	2.308, 2.277, 2.264

Binder crossings for $L = 20/30$ and $30/50$. Autoencoder trained at $T \leq 1.7$, for $L = 20, 30, 50$. With PCA at $L = 30$, $|w^{(1)}|/\sqrt{N}$ and $|m|$ correlate at 0.99996.

Summary of the J_1–J_2 and XY Results

- **J_1–J_2:** the magnetisation misses the stripe phase; PCA finds both order parameters; autoencoders trained on three corners recover the phase diagram, a coarse scan at 0.5% of the cost of a dense one
- **2D XY:** no local order parameter; PCA and autoencoders on snapshots find the magnetisation crossover or the specific-heat maximum, not T_{BKT}; only the stiffness locates the transition
- the method finds where something changes; which order sets in there is the physicist's question

J_1–J_2, $L = 32$	snapshots
dense scan	639 900
coarse + refined scan	3260

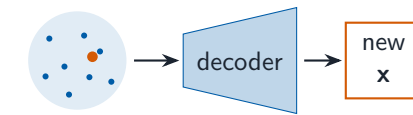
The PCA directions $\boldsymbol{\lambda}_{1,2,3}$ carry 16.6%, 8.4%, 8.2% of the variance

2D XY, $L = 48$	temperature
T_{BKT} (Hasenbusch)	0.8929
stiffness crossing	0.917
specific-heat peak	1.06
PCA, AE on spins	1.03, 1.03

Next Lecture: Learning a Probability Distribution from Data

The data $\mathbf{x}_1, \ldots, \mathbf{x}_n$ are samples from an unknown distribution $p(\mathbf{x})$. In the next lecture we learn p itself and draw *new* samples from it.

- maximum likelihood, and latent-variable models
- variational autoencoders: new digits from random latent vectors
- conditional generation: a chosen digit, or a spin configuration at a chosen temperature



A latent-variable model samples in two steps, first $\mathbf{z}$ from a simple distribution, then $\mathbf{x}$ from the decoder.

For the Ising model we know the true distribution, $p(\mathbf{s}) = e^{-H(\mathbf{s})/T}/Z$: a test of what a network has learned.

Thank You!

Questions, project ideas, feedback: marcin.plodzien@uj.edu.pl

Website · GitHub · Google Scholar

